Princess Sumaya University for Technology
جامعة الأميرة سميّة للتكنولوجيا

KARAM

Karam

A Food-Rescue App Solution

King Talal School of Business Technology

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0.1 Committee Decision: Approval Form Signature Page

0.2 Acknowledgement

0.3 Abstract

The purpose of this project is to explore an untouched market segment in Jordan's food industry, and to create awareness in the consumer. Most Jordanians face difficulties with food affordability due to a multitude of reasons; Political and Economic factors dating past 2020, namely COVID-19, have heavily influenced food supply, affordability, and in some instances, job security. A reported rise in food prices superseding the income growth of Jordanians which has placed a burden on household budgeting and a limit on access to nourishing food. The growing restaurant industry in Jordan indicates a growing desire for dining-out, possibly due to the varying interchangeable lifestyles that exist and a seasonal influx of tourism. Through our application, this project hopes to instill a new-founded awareness that allows for business-owners and consumers to be benefited. The project is an accordance with the Software Development Life Cycle (SDLC) model, with an emphasis on supply chain optimization starting with identifying the opportunity of the untouched segment and collecting the necessary information from both parties (businesses and consumers) through a survey conducted.

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Chapter 1

Introduction

Bridging the gap Food consumption: The food sector in Jordan is a distinct market with plenty of interesting prospects and difficulties. Although the number of restaurants in Jordan is increasing due to the increased demand for a variety of eating experiences, many Jordanians still struggle to make ends meet. A report by the Carnegie Endowment for International Peace (2021), a multitude of intricate variables, such as political and economic shifts that have affected the availability, price, and stability of jobs as well as the food supply, are to blame for this battle, especially in the wake of the COVID-19 pandemic. As a result, families all around Jordan frequently struggle to obtain wholesome meals.

There is a need for convenient and reasonably priced meal options despite these obstacles. This project seeks to close this market gap by putting forth a cutting-edge strategy that is advantageous to both companies and customers. We want to enable companies to cut waste, engage with cost-conscious consumers, and promote a more sustainable food ecosystem in Jordan by creating a novel food application.

This project investigates the potential solutions provided by our proposed application, goes further into the particular issues that people and enterprises in the Jordanian food industry confront, and lays out the steps involved in implementing this novel approach. We intend to open the door for a more accessible, sustainable, and inclusive food experience for all Jordanians by tackling these important concerns.

1 Problem Statement

30% of the food produced for human use is discarded, causing a worldwide surge of food waste. This leads to negative impact on the environment, in addition to tremendous financial losses in the food industry. Many people, including those of Jordan, do not find it easy to obtain reasonably priced, healthful meals. An apparent void is filled in Jordan through an alternate food app that emphasizes sustainability and cost. The Food industry creates 34% waste of its own supply, taxing our environment and available resources, while many Jordanians currently face difficulties providing food to their families. Our app approaches these 2 issues directly. We suggest achieving that by cutting waste and providing excellent, low-cost meals by merging food-industry retailers and budget-conscious customers together on one platform. This innovative solution is ideal for Jordan's needs and market, since it encourages ingenuity, affordability, and environmental responsibility.

2 Background Review

The restaurant industry plays a pivotal role in Jordan's economic prosperity, making up a substantial share of Jordan's GDP; 14.6% of Jordan's GDP comes from tourism, according to the Jordan Tourism Board, where the hospitality and restaurant industry meet (Jordan Tourism Board, 2023). In-spite of fluctuating challenges, this industry remains to evolve and develop; the Jordanian Union of Restaurants and Sweets Shops Owners (JURCP) roughly calculates that Jordanians' expenditure on restaurants is over JD 1 billion annually (JURCP, 2023), translating to roughly JD 90 million per month. Moreover, the number of restaurants, cafes, and sweets shops in Jordan currently stands at nearly 15,000 (JURCP, 2023). From this, it can be concluded that Jordanians are increasingly eating out, with changing lifestyles contributing to this trend. This reflects a growing demand for diverse dining options.

The COVID-19 pandemic had a significant impact on the industry, with a drop in restaurant sales of an estimated 40% in 2022 (Al-Awwad, Jordan News, 2022). In addition, The market is fiercely competitive, with established players, international chains, and new entrants vying for market share. Increasing food costs, labor shortages, and regulations can pose challenges for restaurant owners, impacting profits (Haddad and Al-Omari, 2023). These issues can be alleviated with the proposed project, as it would create a new segment for existing businesses to create a new stream of revenue, where losses were previously prevalent.

3 Project Significance

"Karam" is a mobile application that links consumers with local merchants and eateries in need of selling their daily pre-packaged/ready-to-eat supply at a lower price. Moreover, we suggest the app provide "surprise bundles" with mystery items at a further discount supplied from eateries, appealing to consumers who are willing to take chances and cut down on waste. This project will be solely focused on building partnerships with businesses to prevent a surplus of food supply to be wasted by creating a platform for consumers.

4 Project Objectives

The objectives of this project are as follows:

- Minimize the amount of food waste generated by businesses.
- Provide a new service for consumers to enjoy.

- Create new business opportunities and ventures for Jordanian eateries.
- Streamline operations of an untouched segment.
- Linking investors with potential opportunities.

5 Project Charter

A document issued by the project sponsor that formally acknowledges the existence of a project. It provides the team, manager, and stakeholders with the necessary details about the project.

Project Title	Karam
Project Manager	Dania Juma
Project Start Date	15/01/2024
Projected Finish Date	20/08/2024
Project Background	The food take-out culture in Jordan
Project Objectives	Attempting to reduce food waste, develop an app that connects customers with local businesses and restaurants, sell inexpensive surprise bundles, provide Jordanian citizens with food at more affordable prices, and reduce the environmental impact of food waste.
Success Criteria	The project should meet stakeholders' expectations and needs, be completed and finished in accordance to the project time-frame, and the project budget shouldn't be exceeded.
Approach	The "Karam" application was developed using the Software Development Life Cycle (SDLC) as a template, following the Agile framework for flexibility and efficiency. Key project elements like objectives, scope, deliverables, and milestones were carefully planned, while risk management and quality assurance were emphasized throughout. Built using Flutter, the app was designed to run and be tested on iOS, providing a user-friendly interface while minimizing risks and delivering high-quality software.
Project Scope	The app features location-enabled capabilities for finding surplus pre-packaged food in the area and detailed listings of extra food products with descriptions and photos. It merges multiple varieties and combinations of food, sends out push notifications about new offers and promotions, and incorporates user profiles with order histories. Additionally, an admin panel is provided for managing restaurant and eatery listings, alongside tools for promotional and marketing initiatives.
Success Factors	High user engagement, collaborations with retailers and high on-boarding rate, high-quality customer service, to address problems quickly and maintain high user ratings, security algorithms to maintain user privacy, customization features for users, "favorites" tab, and renewed recommendation engines.
Name and Signature	Anmar Rawashdeh, Team Member, Data Analyst, Anm20200074@std.psut.edu.jo
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Table 1.1: Project Overview: Karam

6 Gantt Chart

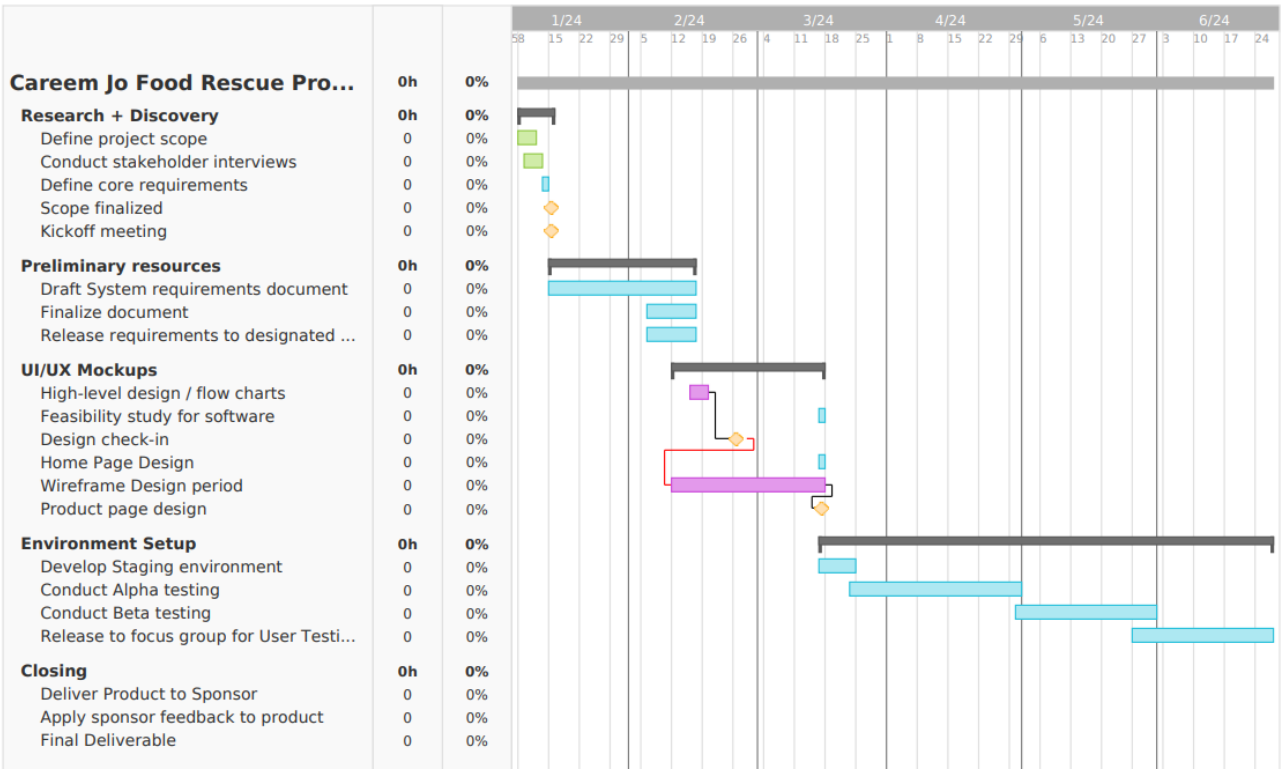


Figure 1.1: Gantt Chart

Chapter 2

Theoretical Review

To accurately asses this project's applicability, competitive analysis, Porter's five forces model analysis, ethical aspects, existing initiatives, and theoretical research research were conducted to study proposing it to the Jordanian market and the surrounding region. This section encompasses critical analysis to understand the relevant aspects and components of this application. A holistic overview is essential in creating a comprehensive framework for planning and navigating business plan deployment. Environmental scanning techniques help marketers identify all aspects of an idea across all industries.

1 Theoretical Background

Approximately 1.7 million people in Jordan are suffering from critical food shortages. This figure constitutes 17% of the population, show the latest figures of the Food and Agriculture Organization (FAO) between 2019-2021; placing the population of Jordan at 11.3 million people, as Saraya News reports (Jordan News 2023). A report issued by Horago, online food orders through mobile food delivery apps produce 23% additional revenue for restaurants on average compared to regular orders.

The National Food Security Strategy (2023–2030) in Jordan showed that more than 50% of Jordanian households suffer from food insecurity, with rates as high as 20% in some areas. Syrian refugees in Jordan are particularly affected, with 23.4% experiencing food insecurity and 60% at risk (Jordan News, 2023). Jordan's large refugee population, primarily from Syria, puts additional pressure on food resources. According to the World Food Programme (WFP, 2024), 53% of refugees are food insecure, many refugees struggle with food insecurity due to limited job opportunities and a lack of financial resources . Challenges contributing to food insecurity include urban expansion, fluctuating food prices, climate change, refugee influx, declining remittances, and COVID-19 impacts. Jordan also grapples with various forms of malnutrition, including deficiencies, obesity, and related non-communicable diseases, due to changes in dietary habits and lifestyles.

Environmental Impact of Food Waste: An estimated 1.3 billion tonnes of food waste is globally produced, as reported by the Food and Agriculture Organization (FAO, 2023). This figure is almost one-third of the food produced for human consumption. Wasted food decomposing in landfills produces methane, a greenhouse gas contributing to climate change. Food waste is considered wasted resources used in production, including land, water, and fertilizers. Moreover, the production of food sometimes leads to irreparable environmental damage like deforestation and habitat destruction, and wasted food exacerbates this strain.

Economic Impact of Food Waste: In Jordan, food waste is estimated at 93 kg per person annually, translating to wasted food that could feed 1.5 million people for a year (United Nations in Jordan, 2021). The FAO estimated the cost of food waste to be around 800 USD billion annually, which could feed around 811 million hungry people globally. Resources used in food production that ultimately goes to waste strain the global food supply chain, the income stream of farmers, and food security issues.

Gaps in Current Food Shortage Solutions in Jordan: Existing initiatives like food banks and donation programs (expanded more on in **Competitive Analysis**) often struggle to reach all vulnerable populations due to logistical constraints and limited storage capacity [1]. Matching supply (surplus food) with demand (individuals in need) can be inefficient, leading to food spoilage and missed opportunities. Consumers might be unaware of the availability and location of surplus food offered through traditional channels.

Benefits of Mobile Applications for Food Accessibility: Applications hold the power to connect businesses with a wider consumer base, ensuring surplus food reaches those in need. Apps can provide real-time updates on the availability and location of discounted food items. Applications facilitate a direct connection between food sources and consumers, minimizing logistical hurdles. Consumers can easily browse available options and access discounted food through their mobile devices.

Impact on Consumer Behavior: Apps can encourage consumers to reduce food waste by offering a solution by utilizing surplus food. Discounted food options can improve food security for individuals with limited financial resources. By reducing food waste, applications contribute to a more sustainable food system.

- *Increased Value Consciousness:* The app facilitates an environment of mindfulness and value that increases awareness towards consumption.

Limited Household Income: Food insecurity is not evenly distributed across Jordan. According to a report by the Carnegie Endowment for International Peace, rural areas tend to be worse off, with Al-Tafilah governorate having the highest percentage of food insecure households. Food prices in Jordan rose 1.2% by end of August as reported by the World Bank in 2023 (World Bank, 2023). Economic factors like slow growth and unemployment contribute to a situation where many Jordanians simply don't have enough income to cover rising food costs. This forces them to choose cheaper, less nutritious options (Borgen Project, 2019).

2 Competitive Analysis

2.1 Existing Initiatives to Address Food Waste in Jordan

Food Banks: Organizations like the Jordan Food Bank collect and distribute surplus food to those in need (Roya News, 2021).

Limitation: Limited storage and transportation capacity restricts the amount of food they can handle.

Donation Programs: Restaurants and supermarkets may donate nearing-expiry food to charities (The Jordan Times, 2023).

Limitation: Logistical challenges and potential food safety concerns can hinder participation.

Composting: Organic waste, including food scraps, can be composted to create nutrient-rich soil.

Limitation: Requires infrastructure development, proper waste separation at source, and knowledge for effective implementation.

The National Food Security Strategy (2021-2030): This national framework prioritizes initiatives to reduce food waste across the food supply chain (Jordan. National Center for Supply Chain Management [JNCSCM], 2021). The Ministry of Agriculture's "No Food Waste" Initiative: Launched in 2022, this initiative aims to:

Raise Public Awareness: Educational campaigns target consumers, businesses, and institutions to promote responsible food practices. The Ministry collaborates with existing organizations working on food waste reduction.

A Collaboration with International Organizations: The Ministry of Agriculture works with the Food and Agriculture Organization (FAO) and the World Food Program (WFP) on initiatives like:

- Formative research: Studying consumption habits to identify areas for reducing food waste at the consumer level (United Nations in Jordan, 2021).
- Enhancing food waste management: Supporting local organizations involved in food recovery and redistribution.
- The Jordanian Ministry of Agriculture announced a plan to establish 15 new food industry factories by the end of 2023 (Jordan Times, 2023).

2.2 Global Competitors

Similar iterations of this concept have been implemented, namely, Too Good To Go, as it is a successful example in European and North American cities. Munch is a similar application in Hungary, and Karma in Sweden, which all serve as practically useful in building this business model. From a competitive viewpoint, the above examples are not a threat as they are not focused on the Jordanian and MENA region. While not solely focused on food rescue, large food delivery companies like Uber Eats and DoorDash, (Careem and Talabat in Jordan) could be perceived as indirect competitors. They offer similar convenience to consumers and might partner with stores to manage surplus food, potentially impacting the pool of donations available to the app. More is expanded upon in **Chapter 3: Competitor Functionalities**.

3 Proposed System

Overcoming food insecurity and the solutions to mitigate it are essential in protecting and improving the livelihood of Jordanians struggling with this issue. This proposed project targets a large faction of Jordan's population, namely young working adults, extending into struggling families looking for healthy, accessible, and affordable meals. Environmentally-conscious consumers are the secondary demographic targeted.

Consumers can enjoy price-reduced food, saving their money on other priorities such as groceries. This enables them to explore culinary possibilities provided by local eateries, which boosts their economic livelihood. An added benefit is mitigating a business's ecological impact by avoiding food wastage. Reducing food waste and associated disposal expenses, increased income by offering to sell extra food at a reduced price, and the app's marketing tools to draw in new users and consumers are the main value propositions offered by this platform for local eateries and restaurants.

Upon downloading the application, the user is greeted with a registration page to create an account. The user will have a choice to add a credit card for online payment, which is the primary recommended option as the application works on a first-come first-serve basis. Through GPS-enabled location services, the user will be able to start browsing nearby restaurants and options, and choosing from categories; snacks, bakeries, pastries, meals, and groceries. Through user-enabled push notifications would be notified when locations nearby have food ready to be rescued. Users then pick a suitable time for pick-up.

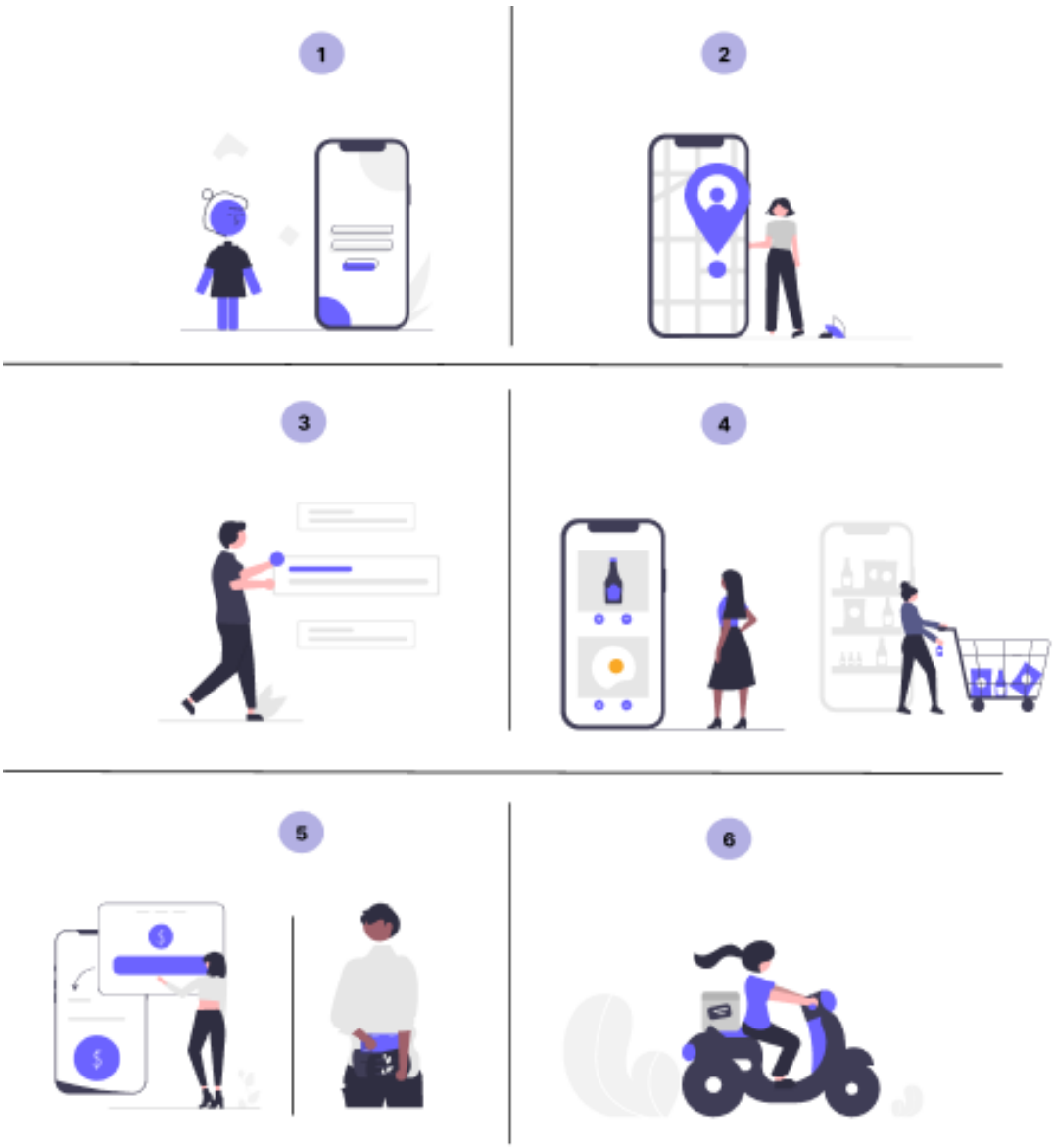


Figure 2.1: Proposed System Model

3.1 Ethical Considerations

Our mission is to inspire and encourage people in Jordan to stop food waste, when businesses like restaurants, bakeries or cafes have a surplus of food by the end of the day due to a number of factors, they list it on our app. Customers will then browse our app for the surplus food that those business have listed, and then they'll buy it for a discounted price, and either pick it up or have it delivered to them for them to enjoy. This way the businesses won't suffer from financial loss, the food isn't wasted and customers get delicious food for an amazing price.

By selling surplus food for discounted prices, we'll also be helping financially limited people, such as university students, have the ability to eat good food for a price they can afford without having to worry about spending too much, and the best part about it, is while they're doing

that, they'll be contributing to stopping food waste.

Stopping food waste is not only helpful for the business and customers, it is also helpful for our environment. When food ends up in landfills, it dose not decompose without harm, due to the lack of sufficient airflow in landfills, as the food is decomposing it releases methane, which is a greenhouse gas that is 30 time more harmful than carbon dioxide, over the course of 100 years, which means that food waste is one of the leading causes of climate change. A major ethical concern is food safety and regulation. Quality Assurance is vital, whether it is internally or externally appointed (can be government-appointed upon project approval, or internally through the application personnel or a local eateries' representative. This is a top priority to ensure consumers feel confident in the transparency and reliability of this project. A first-come first-serve approach would eliminate exploitation and over-consumption of the concept, and provide equity among users.

3.2 Legal and Regulatory Aspects in Jordan:

It is crucial to consult with relevant authorities like the Jordan Food and Drug Administration (JFDA) to ensure the mobile application adheres to all food safety and handling regulations; compliance with legal food licensing and regulations, in regards to food handling, storage, and final sales. This would entail obtaining the required permits and health code requirements, establishing clear food safety protocols, where explicit guidelines and criteria are set in place for accepting food donations (immediacy of expiry date, and proper storage conditions). Establishing the necessary clear terms and conditions that users agree to upon creating an account is a primary step in guaranteeing safe conduct in legalities for unseen circumstances. Financially, settling taxation implications for in-app transactions is another consideration. Food Safety and Handling: Jordan has established regulations outlined in the Jordanian Food and Drug Law No. 31. These regulations ensure the safety and quality of food throughout the supply chain. Applicability to Mobile Solutions: Businesses utilizing applications for food distribution need to comply with these regulations. This might involve: *Traceability:* Maintaining records of food origin and ensuring proper handling procedures. *Temperature Control:* Implementing measures to maintain appropriate temperatures for perishable food items. *Licensing:* Obtaining necessary licenses for food handling and distribution activities.

Collaboration: Partnerships with government agencies, NGOs, and food businesses can strengthen the reach and effectiveness of the application. Public Awareness: Educational campaigns can raise awareness about food waste reduction and encourage consumer adoption of

the application.

By addressing these gaps and adhering to legal requirements, mobile applications can play a significant role in improving food accessibility, promoting responsible consumption habits, and contributing to a more sustainable food system in Jordan.

Best Practices:

- Consulting with legal council specialized in food safety and business law in Jordan to develop the required best practices in the development in this project throughout each phase of development is essential.
- Additionally, collaborating with organizations specialized in food insecurity and safety to ensure compliance with standards.
- Clearly communicate the application's mission and purpose, highlighting the intent on mitigating food waste and providing affordable food.
- Consult with legal representatives about User's rights to gain and enable location services via the app, and acquire their location with User permission.

4 SWOT Analysis

Strengths:

- *Reduced Food Waste:* Through giving restaurants and bakery's a platform to solve a major issue of surplus and helps these businesses sell extra food / baked goods before they go bad and end up being wasted.
- *Cost Effective:* Offers customers with a time limited purchase at a discounted price, making it appealing to those who are willing to pay for the price.
- *Convenience:* Gives access for customers to access discounted goods through a mobile app, saving the customer the time and effort for searching for any deals or alternatives.
- *Convenience:* Gives access for customers to access discounted goods through a mobile app, saving the customer the time and effort for searching for any deals or alternatives.
- *Helps Local Businesses:* Enhancing customer circulation and revenue for nearby restaurants and cafes, through the promotion and building a customer base that supports the environmentally business practices.

Weaknesses:

- *Quality Concerns:* There may be a couple of concerns from consumers regarding the quality and how fresh the food could turn out to be, which can potentially dissatisfy customers.
- *Dependence on participating businesses:* The success of the app heavily relies on the bakeries and cafes to participate on this program and to give in all they can as they have a major role to pay off.

Opportunities:

- *Partnership and Collaborations:* Establish connections with food producers, retailers, and eateries to expand your product line and attract new customers.
- *Geographical Expansion:* Scale the app to new regions or cities, increasing the market of the catering service to the needs of diverse populations.

Threats:

- *Creation:* Face competition from existing apps such as Talabat and Careem as they provide a wide variety of services and many discount offers that may have similar services.
- *Consumer Behavior:* Adoption rates and usage patterns of the app may be impacted by changes in consumer preferences or behaviors, such as heightened worries about food safety and hygiene in the wake of public health emergencies.

5 Porter's Five Forces

Threat of New Entrants: *Low:* Although creating a smartphone app may not require much in the way of finance or technological restrictions, forming partnerships with bakeries and cafes and growing a user base may be difficult. The rivalry will likely increase if already-existing meal delivery platforms or apps enter this market and take use of their user base and infrastructure.

Bargaining Power of Suppliers: *Medium:* The availability of excess baked products from vendors is a determining factor in the app's success. Bakeries and cafes can negotiate better prices, particularly if they have other outlets for selling their extra stock. On the other hand, by providing a platform to lower food waste and draw in more users, the app can bargain

for advantageous terms.

Bargaining Power of Buyers: *High:* Because costumers can quickly switch between several meal delivery apps or buy baked products from other suppliers, consumers have a lot of negotiating leverage so it's best if we best meet their demands. The availability of alternatives and price sensitivity may have an impact on their purchase decisions. But this danger might be lessened if the app offers distinctive value propositions like ease and lowered costs.

Threat of Substitutes: *Low-Medium:* The threat of substitutes is minimal because there are not many places where people may get baked products besides grocery stores, bakeries, cafes, and other food delivery services. But the app can set itself apart by allocating to decreasing food waste and providing special pricing, which might lessen the risk of replacements.

Industry Rivalry: *Low-Medium:* Regionally, there are limited concepts similar to the proposed concept, providing a competitive advantage in and beyond the Jordanian audience. The direct competitors do not provide the uniqueness and financial advantage that this project provides.

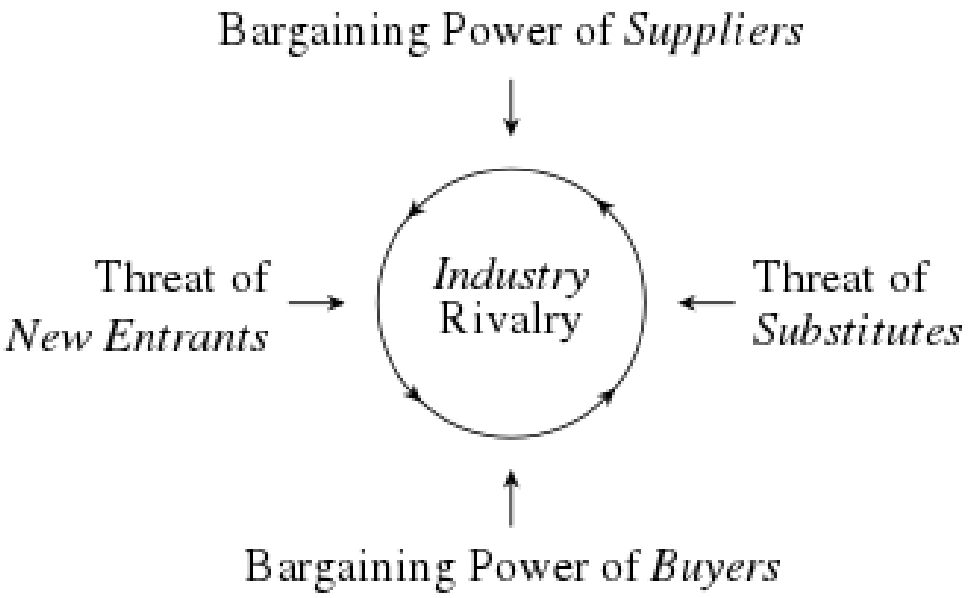


Figure 2.2: Porter's Five Forces Model

6 Scenario Analysis

This analysis explores potential scenarios for an app that connects restaurants with customers seeking discounted surplus food on a first-come, first-served basis.

Scenario 1

- *Scenario 1: High Demand, Successful Launch:* The app receives significant user interest and quickly attains traction with both target users, restaurants and customers. Users would appreciate the deals and contributing to reducing food waste, while restaurants profit from from utilizing potential waste of good product while generating additional revenue.

- **Best Practices for Scenario 1:** Well-structured marketing campaigns aimed at both restaurants and budget-aware customers, a User-friendly interface focused on understandable information about the available items, discounts and promotions, and GPS-enabled pickup locations. Efficient order fulfillment and order pick-up require a comprehensive logistics system.

- **Challenges:** Managing high demand: The application could struggle to meet surges in users and orders, while ensuring enough surplus food is available; Scalability: It is essential to build adaptability to accommodate a growing user-base and restaurant network; Food Safety: Clear guidelines and protocols are an essential criterion to guarantee food quality and prevent spoilage.

Scenario 2

- *Scenario 2: Low User Adoption* The app fails to attract enough users, leading to limited participation from restaurants.

- **Challenges:** Insufficient marketing tactics and decreased user attraction/targeting the wrong audience, prices may not be low enough to incentives users, inconvenient pick-up locations or times; limited pick-up options may deter users.

- **Solutions:** Focus on reaching budget-conscious customers and environmentally-conscious demographics; Negotiate better discounts with restaurants; A higher discount would encourage users; Partner with delivery services or offer extended pick-up windows.

Scenario 3:

- *Scenario 3: Food Quality Concerns.* Customers have negative experiences with food quality, leading to bad reviews and user churn.
- **Causes:** Unclear communication about food expiration - Users might be unsure about the freshness of discounted items. Lack of quality control - Restaurants may unknowingly offer spoiled food.
- **Preventative measures:** Implement a clear food labeling system - Highlight the "best by" date and any potential imperfections. Partner with restaurants with a strong reputation for food safety. Offer a refund or replacement policy for unsatisfactory food.

Scenario 4

- *Scenario 4: Competition from Established Players.* Existing food delivery apps or grocery stores launch similar features to combat food waste.
- **Competitive advantage:** Focus on niche market; Partner with local restaurants to offer unique and diverse food options; Promote the app's role in reducing food waste and fostering social responsibility.

Chapter 3

Business Requirements

It is essential for a system to complete a list of requirements (human, functional, and non-functional requirements, and a feasibility study) to create a comprehensive framework of the project understanding. Determining user needs and requirements is a process led by the user, implemented by app's developers and team.

1 Human Requirements

Application Developers:

- Developers: build and maintain the app.
- UX/UI Developers: build the interface and front-end portion of the application.
- Content Creators: manage the application's information and promotions.

External/Business:

- Business Partnerships division: responsible for establishing partnerships with local eateries and restaurants.
- Food regulation and safety professionals: to ensure the proper protocols and guidelines for handling and storage.
- IT Project Manager: ensuring effective project execution.

Outreach:

- Outreach team: a team responsible for collaborating with charities and shelters for holistic food distribution.
- Includes specialists that raise awareness about the application's purpose and benefits.

Customer Service Representatives:

- Technical team: staff that manage operational issues; B2C.
- Call Center: Provide support for app users and troubleshoot any issues.

Operational:

- Operations Manager: oversee and coordinate daily operations, manage staff, and ensure overall app functionality from the operations department.

- Third-Party Delivery Contractors

Management:

- Finance: Maintain financial health, reporting, investment analysis, and risk management.
- HR: recruitment and on-boarding, compensation and benefits, performance management, training and development, and employee relations.
- Research and Development: innovation and product development, market research, prototyping and testing, technology scouting, and intellectual property preservation.

2 Functional Requirements

Since Karam revolves around food waste, it is essential to asses the awareness and level of care, and aim to create a better culture around commercial food waste. Food price inflation exceeded overall inflation in 48.8% of 166 countries where data is available; Global food price inflation has placed a strain on common households, threatening the procurement of proper nutrition ("Food Security Update," 2024). Functional requirements outline what functionalities are necessary for a system to perform optimally. The below text provides a breakdown of the functional necessities for Karam, extremely important to convey to stakeholders what the system offers.

General Functionalities:

1. *User Profiles:* Profiles that allow users to save preferences, payment info, and order history.
2. *Review System:* A system where users can write reviews for restaurants.
3. *Chat and communication tools* that users, delivery drivers, and staff can use for troubleshooting.
4. *Notifications and Alerts:* The app should send notifications to users when new items nearby are available, and to general promotions and new restaurant additions.
5. *Cloud Integration:* The app should store information relevant to the user's actions, preferences, and history.
6. *User Interface:* A user-friendly interface focused on prioritizing user enjoyment.

User-Centric Functionalities:

GPS Tracking

1. Map interface providing a visual interactive page to browse options and restaurants.
2. The app should be able to accurately track a user's location, and provide accurate location services to view restaurants and eateries to avoid pick-up confusion.
3. Location services should be able to create push notifications about close-by available items.

Filtering System:

- Dietary restrictions and food preference to help tailor search results.

- Browsing categories and discover page; comprehensive smart filtering categorization system.

Payment and Handling:

- Ordering and payment gateway, where users can safely proceed with secure payment options.

Loyalty Program:

- A point system based on orders, to reward and encourage continued use. This not only retains current users, but encourages new users through referral points and cashback discounts.

Social Media Integration:

- Customers can use social media integration to easily share photos/videos, which are linked to their social profiles with log-in capabilities.

Discounts and Allowances:

- The application should offer discounts and promotions to users, such as free delivery/discount codes.

Business-Centric Functionalities:

1. Businesses and eateries require a detailed platform to build their profile on, receive and coordinate orders, and update their listings.

2. Dashboard Services: Businesses and the application's staff require a dashboard mechanism to be able to track sales and expenses (individually-tailored for a business, and a larger full-encompassing of all businesses for the application.

3. Menu Management System: The ability of restaurant owners to easily navigate their items.

3 Non-Functional Requirements

Non-functional requirements place a criteria that specify standards about the system's operations and qualities. Non-functional requirements (NFRs) in software development are criteria ensuring that a product meets business requirements effectively. While functional requirements dictate what the software should do, NFRs focus on how well it should perform, encompassing aspects like speed, compatibility, and localization (GlobalLogic, 2023).

1. *Compatibility:* The app should be compatible with a variety of mobile devices.
2. *Serviceability:* The application should be centered around usability, with consideration to factors like ease of use and simplicity.
3. *Handling of Errors:* Ensuring implementing a error-mediating and handling mechanisms to face malfunctions/failures.
4. *Compliance to Regulations:* The application must comply with Jordan's data protection, preservation, and privacy maintenance.
5. *Security:* All data collected in the app through the user (financial information, personal information) is encrypted and secure.
6. *Performance:* Low-to-little latency in transmitting data to point-of-sale systems that alert restaurants of incoming pick-ups/delivery.

3.1 Competitor Functionalities

Food-rescue apps like FlashFood, Too Good To Go, Karma, and Munch all serve as platforms to connect businesses to users, selling surplus food at discounted prices. Their functionalities include benefiting businesses by recouping lost costs, benefiting consumers by providing access to discounted meals and reductions in grocery bills, promoting sustainability, and reducing general food waste.

- *Too Good to Go:* Users view close-by restaurants/grocery shops/bakeries and shop for "surprise bundles" that contain a variety of items from an eatery/restaurant. The bundles are picked up by the clients at a specified time.
- *Karma:* Users choose from a variety of categories of popular/near-by categories, from a selection of restaurants and grocery stores with a focus on the retail level.

- *Munch:* Munch users can either shop from the discounted section or the company's dedicated market with discounted house essentials.
- *Flashfood:* Produce, meat, and more house essentials that are near expiry/leftover inventory are discounted half-off. The Flashfood app advertises deals and discounts for local grocery stores that would appeal to a large group of people.

3.2 Indirect Competitor Functionalities

Careem and Talabat :

1. Extensive Restaurant Selection: Both apps have a wide variety of restaurants, that provides a broad range of cuisines for the customers cravings.
2. Convenient Browsing: To come up with a well informed judgments on your dinner, a customer can browse the menus, visually appealing food photographs and customer reviews.
3. Customization Options: A customer can feel free to personalize their orders with special interactions for restaurants, to ensure that the customer is fulfilled with their dietary needs and preferences.
4. Secure Payment Methods: A customer may safely pay for their meals using debit cards, credits cards or mobile wallets like Apple pay, google pay or Samsung pay.
5. Delivery Tracking: The customer is sent push-notifications and can view their delivery driver in real-time via GPS-enabled live-tracking.
6. Real- Time Order Tracking: A customer can monitor the progress of their order in real time, providing a estimated time frame and allowing the customer to track their delivery though the location on the app.
7. Transparency Throughout Delivery : Knowing when to anticipate your food to come can ease your worries.
8. Exclusive Deals and Discounts: A big part in both applications is that they take a big advantage in giving special offers, coupons and restaurant promotions which helps to save money on the customer.
9. Subscription Plans: Both platforms provide membership plans to its regular customers, which may include privileges to the customers like free delivery after a specific amount of purchases placed.

10. Order Scheduling: Helps the customers order in advance for a later delivery time to ensure that the food arrives exactly on the time needed.

Choosing Between Careem and Talabat:

1. Restaurant Availability : A customer can easily check both platforms' offers and choose the specific restaurants that they are interested in.

2. Delivery Areas: Each app has ensured that restaurants can delivery to certain locations.

3. Promotions and Discounts: A customer can compare the deals and coupons available on each app.

4. Personal Preferences: Both consider huge factors such as User interface/UX and ease of use.

4 Feasibility Analysis

A feasibility study is an in-depth study and analysis of all the aspects that critically contribute to the aspects of a proposed project with the goal of assessing the project's chances of success. Return on investment, or the project's ability to create a profit big enough to make up for the investments made, is also a main indicator of a successful business venture. However, many indicators can be identified as positive or negative and many several aspects, such as the influence on the environment and community responses that may be observed.
Project managers can use feasibility studies as a means of help to evaluate the many risks and returns of a suggested plan of action, but before anything is green-lit a couple of things need to be put into consideration.

4.1 Operational Feasibility Study:

One of the major points regarding the project is to conduct means to evaluate the project using operation feasibility studies as it helps us with the understanding whether the project system is worth implementing from an operational perspective. This kind of research evaluates several variables, including the organization's capacity to accept and maintain the new project or system, its compatibility with current systems, its possible influence on operations and the availability of resources. Implementing an operational feasibility study plays a huge factor in our project as the compatibility and ease of use of our application of users can determine a huge role of success. One of our most crucial requirements in implementing this project is to make the application stand out with its easy user-friendliness as it's the most critical part of our operation. As the application is meant to be targeted to a huge audience in and outside Amman for easy drop in goods at a cheaper price. For that reason, our main objective is to prioritize the user experience and interface. As Karam is an application research project, developing the application takes a lot of time and effort. Nonetheless, we intend to engage the services of the app development team to provide a fully functional application. The development team will consist of a Full team of developers and IT managers.

4.2 Technical Feasibility Study:

Creating a smartphone application for restaurants to display their goods seems like a promising technological endeavor. Due to the increasing use of smartphones and developments in mobile technology, a large range of frameworks and development tools are available to create high-

caliber mobile apps. We could easily construct a single code-base that runs on Android devices by utilizing cross-platform development frameworks using Flutter, which would cut down on development time and expenses. Furthermore, including functions like menu browsing, online ordering, and accurate destinations on google maps into the mobile application can improve user interaction and expedite the ordering process. *The Requirements are:*

- Powerful computers and a setup (To operate on the mobile application)
- Multiple Smartphones (Prototyping for multiple operating systems)
- Social Media account(s) (For Marketing purposes)
- Creating a Wireframe for the Application (Designing the application)
- Flutter Flow (Developing the Application)

All these requirements will allow "Karam" to fully operate in sync.

4.3 Financial Feasibility Study

This section assesses the financial feasibility of Karam in Jordan that connects users with discounted unsold food from eateries, cafes, and bakeries. Creating a financial framework is essential for assessing a project's financial feasibility; predicting costs, revenue, and return on investment (ROI).

Problem: Food waste is a significant issue globally, with Jordan estimating 93 kg per person wasted annually (United Nations in Jordan, 2021). Food waste is a global problem with environmental and economic consequences.

Target Market:

- Budget-conscious consumers seeking affordable meals.
- Environmentally conscious individuals seeking to reduce food waste.
- Busy professionals seeking convenient meal options.

Fixed Costs:

Operating Expenses	**Cost Per Month**	**Costs Per Year**
Quality Assurance Officer	700 JOD	8,400 JOD
Security Manager	1,200 JOD	14,400 JOD
Financial Accountant	1,000 JOD	12,000 JOD
Marketing Team	5,000 JOD	60,000 JOD
Office Spaces	1,000 JOD	12,000 JOD
Utilities	200 JOD	2,400 JOD
Call Center	600 JOD	7,200 JOD
Total	9,700 JOD	116,400 JOD

Table 3.1: Fixed Costs

Start-Up Costs:

Item	***Cost***
Work-Issued Devices (Laptops)	600 x4
Google Maps API (Maps Javascript API)	Free
Application Platform(s)	AppStore $99 Yearly, Playstore: $25 Fee
Developers Team	1,200 JOD x8 months x4 members
Project Manager	2,000 JOD for 8 months
Full-Stack Head Developer	1,500 JOD for 8 months
Registration and governmental Fees	500 JOD
Total	69, 425 JOD

Table 3.2: Start-Up Costs

In building the proper team for app development, included above in the start-up costs table is an estimation for work-issued devices, team salaries, and additional costs that could be incurred. Also included is the consideration of the Google Maps Platform, which is a set of APIs and SDKs that permits developers to embed Google Maps into websites and mobile applications.

Financial Projections:

Revenue:

- Commission on transactions between users and partner businesses (e.g., 10-15%).
- Premium Subscriptions (Optional): Offer an optional premium subscription with features like early access to deals, delivery prioritization, and loyalty rewards. (Subscription revenue will be projected based on market research and industry benchmarks).

. **Costs:**

- App development and maintenance.
- Marketing and user acquisition.
- Payment processing fees.
- Commissions to partner businesses.
- Staff salaries.

Financial Break-Even Point: The point where revenue equals total costs. This will depend on factors like user acquisition costs, transaction volume, and commission rates.

BEP (Transactions) = Total Fixed Costs / (Average Transaction Value * (1 - Commission Rate - Partner Commission Rate)).

Assumptions:

- *Average Transaction Value (ATV):* 10 JOD.
- *Commission Rate:* 15%
- *Partner Commission Rate:* 5%
- *Total Fixed Costs (App Development, Marketing):* 116,400 JOD.

Break-Even-Point (BEP) Calculation: BEP (Transactions) = JD 116,400 / (JD 10 * (1 - 0.15 - 0.05)) = JD 116,400 / (JD 8.5) = **13,695 transactions**.

Investment Requirements: Funding will be required for initial app development, marketing campaigns, and operational expenses. The specific amount will depend on the chosen development approach, marketing strategy, and staffing needs.

Risk Analysis: User adoption and attracting a critical mass of users and partner businesses is crucial. Competition and established delivery apps might offer similar features in the future. Food safety and maintaining strict quality control procedures is essential. Wastage levels and relying on unpredictable food surplus from partners.

Revenue Streams:

One of the many revenue streams that could be generated through the app is commission fees as some money can be earned from a percentage off the food sold through the app, another service that can be added into Karam is a monthly subscription fee that adds a premium service for customers with gained features such as faster delivery times, no advertisements and free delivery. Advertisements can be added onto the app which could provide a higher revenue stream for Karam, as it can help provide a platform for local businesses and economic friendly brands onto the app which can give users a better access to less niche businesses which will initially help for greater customer satisfaction on both ends.

Cost Analysis:

The only concern of when it comes to the technological costs is the app development as it needs the right team and resources when it comes to developing such app, time and cost will be taken from the development team since its needed to develop maintenance and security which both so much costs to ensure that the safety of the app is in check for both the users and Karam developers to sustain customer satisfaction. Another issue when it comes to the reach of this app is the customer acceptability of such services due to the fact that some people might not accept this type of service to be applicable in this culture and to come up with a marketing campaign will take a serious amount of costs and time, as its needed to develop a customer base which will be needed to educate on the matter of food waste and why its needed to be rescued. To sum up, operational costs encompass a crucial amount of functions to be considered as its used to ensure smooth business operations. Staff salaries represent a huge amount of the portion to the costs as they are required to manage the entire operations such

as customer support and business growth strategies. Customer service staff are required to be mentioned as they provide support with the queries, complaints of the customers and provide them with the best experience so their salaries and costs are to be considered. Obtaining office space, equipment, and software licensing for everyday operations. All these costs together are to guarantee that Karam scales to provide a better standard for the app.

Projected Growth:

Projected growth for Karam relies on analyzing customer trends and needs as its very important for the Karam app, due to the growing awareness of food waste concerns and the need for others revenue sources, restaurants are expected to first join in on the platform rapidly. Accelerated enrollment needs to be achieved as it's used to achieve the planned marketing promotions to initiate the partnership within the local businesses. It's very important to track the market to study the growing customer needs to sustain the demands to ensure the adoption of Karam is to be held carefully.

Exit Strategy: Potential acquisition by a larger food delivery platform, expansion into other regional markets.

Additional Considerations:

- **Partnerships:** Collaborate with restaurants, cafes, and bakeries offering high-quality surplus food.
- **Logistics:** Develop an efficient system for collecting and delivering discounted food.
- **Marketing Strategy:** Target social media advertising and partnerships with sustainability-focused organizations.
- **Scalability:** Design the app for future growth in user base and geographic reach.

Chapter 4

Methodology

1 Project Methodology

The Software Development Life Cycle (SDLC) was used as the template in creation of the "Karam" application, as well as the Agile framework. Secondly comes the planning phase by planning out the objectives, scope, deliverables and milestones that should be established for this project, furthermore it is critical to identify the project stakeholders and establish a communication channel, and to finish off with the execution phase it is best to encourage the collaboration and communication with the team in order to address any issues effectively (Javatpoint, n.d.).

One of the main processes within the risk management is to regularly asses risks throughout the project life cycle. While working on the project it is best to maintain quality and to assure that the established quality is within standards and metrics early on as it can show a good sign of pace towards the project, moreover its important to conduct regular reviews,testings and quality checks to ensure the standards of the project. One of the most vital priorities is to implement a user friendly interface so through establishing continuous evaluations such as conducting post-project reviews, in order to meet areas of improvement.

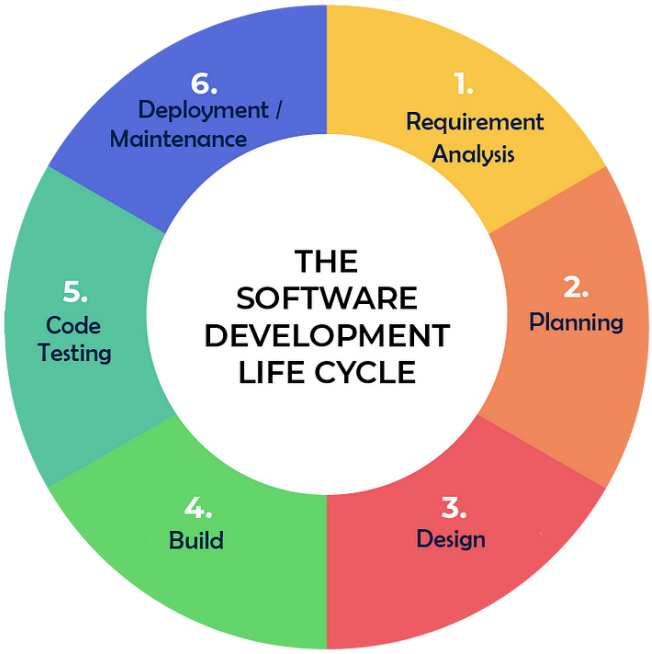


Figure 4.1: Software Development Life Cycle

The framework of Karam is in accordance with the Software Development Life Cycle (SDLC), following a waterfall project management style. High-quality software may be planned, created, tested, and deployed in an organized manner with the help of the software development

life cycle, or SDLC. It offers a road map for the full software development process, guaranteeing that the product fulfills user requirements and performs as planned. The objective is to reduce risks and deliver high-quality software in an efficient manner. The proposed application is built with Flutter to be run and tested on IOS.

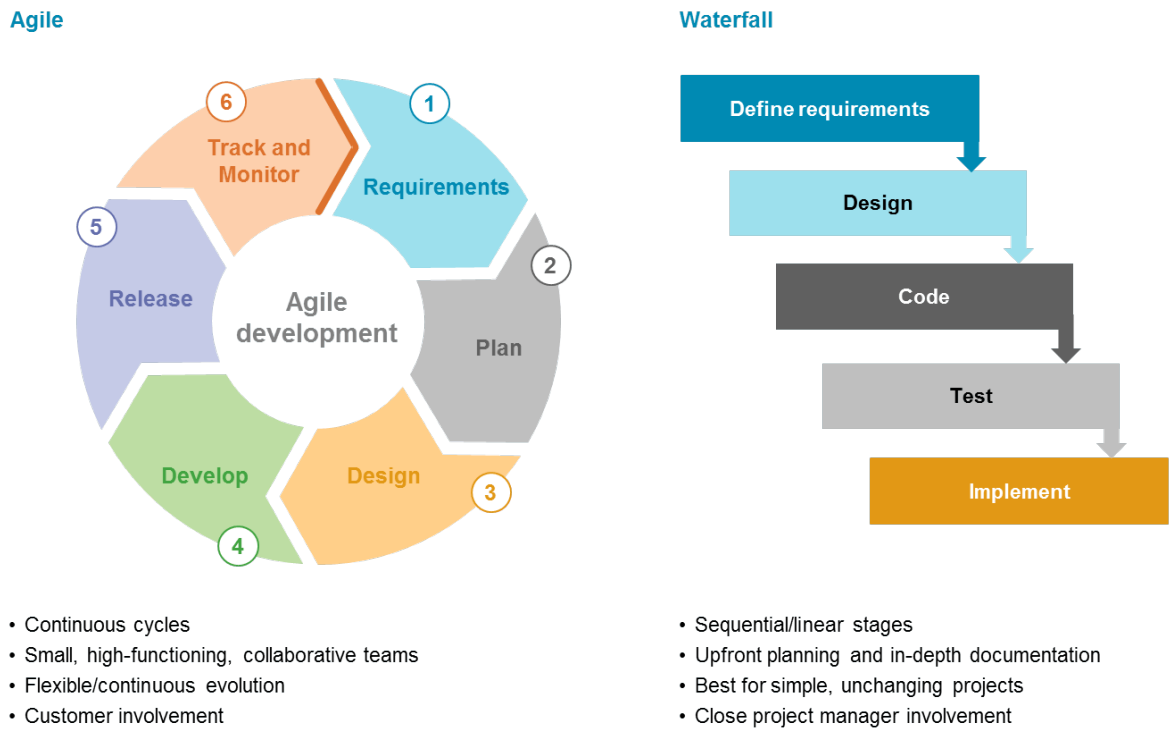


Figure 4.2: Waterfall Project Management

A marketing strategy with a strong foundation is essential in spreading awareness and build a user base. A referral-point based system would be set in place to increase interaction and word-of-mouth advertising/exposure; new users are more motivated to join from a friend's referral. Users are able to enter their dietary requirements, creating a unique and focused experience. This can be achieved through targeted social media campaigns to directly interact with the targeted user demographic, and sponsored advertising. In addition to collaborations with community organizers, which would increase project visibility through content creators' platforms, aiding in credibility and influence.

2 Data Collection and Gathering Techniques

A collection of participants conducted an online questionnaire in order to collect quantitative data. The questionnaire consisted of questions designed to provide insight into the reception of this idea and its features. For this reason, two sections were included to accommodate both users and business owners, to fully gain an understanding of all parties involved. Addition-

ally, personal demographic information such as age, gender, occupation, residence, and their food-ordering behaviors were collected to fully comprehend the multiple perspectives of the participants and potential users.

2.1 Quantitative Data

Quantitative data was collected through an online questionnaire that was shared online. The main areas covered are formed to tackle the acceptance and their perception of Karam:

- **Demographic Information:** Age, gender, occupation, and residence location in Jordan.
- **Consumer Behavior:** Frequency of ordering food from restaurants via mobile applications, awareness and concern about food waste in Jordan, impact of recent food price increases on eating out habits, and importance of saving money on takeout orders.
- **Attitudes and Preferences:** Interest in using an app to purchase discounted, ready-to-go meals, features important in a food waste reduction app, Appeal of using an app to purchase unsold food at a discount, Perception of how well the app addresses the issue of food waste, Likelihood of recommending the app to others, Reasons for hesitation in using such an app, and Incentives that would encourage app usage for reducing food waste.

Quantitative Data Analysis

This section covers the systemic analysis of each question in the questionnaire to extract insight. Google Forms and Excel were used to clean and analyze data collected from a total of 78 responses:

Demographic Information Analysis: Our respondents were a randomly collected sample of the population aged 18 and above. The age groups in the questionnaire were categorized into:

- Under 18
- 18-28
- 29-55
- 55 and Above

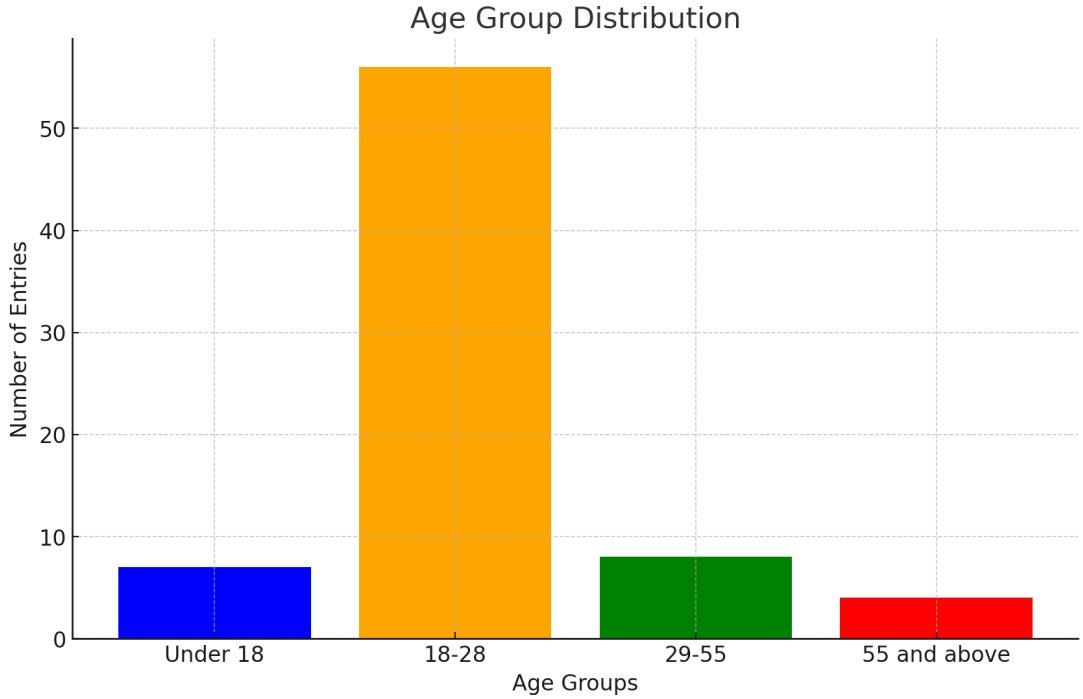


Figure 4.3: Age Distribution

The highest age group percentage was 18-28 at 69.3%. Following it are age groups; Under 18 at 13.3% and 29-55 at 12%, and finally 55 and Above at 5.3%.

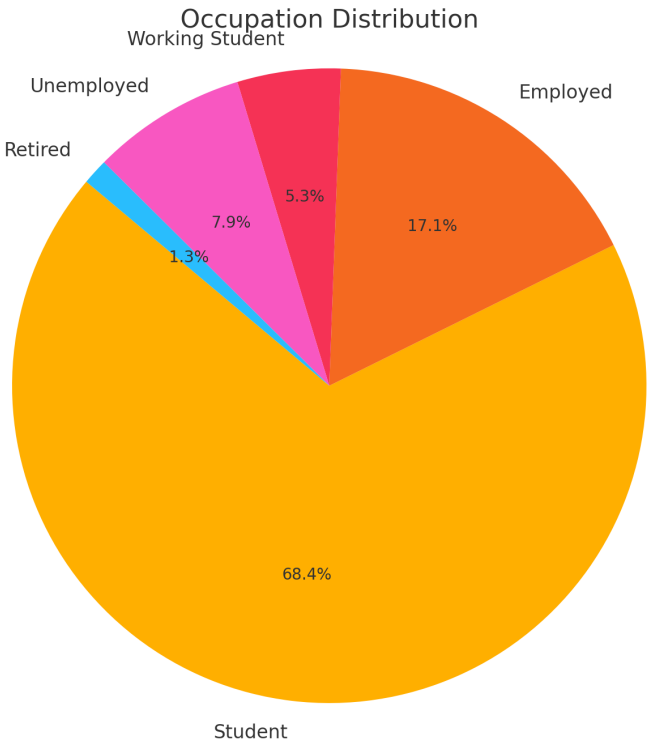


Figure 4.4: Occupation Distribution

To better target and learn about the app consumer audience, the occupation of the respondents were polled; Employed: 17%, Retired: 1%, Student: 69%, Unemployed: 8%, Working Student: 5%.

General Interest and Acceptance of Concept:

"How likely are you to recommend this app to others?"

Figure 4.5: Recommend Survey Question

Analysis:

High Recommendation Potential: A majority of the respondents (50%) rated their likelihood of recommending the app as very likely (rating 5), indicating strong approval and satisfaction with the app's functionality or impact.

Likely to Recommend: A significant number of participants (30.26%) are likely to recommend the app (rating 4), adding to the positive feedback. Neutral Opinion: A smaller portion of respondents (15.79%) are neutral (rating 3), which could indicate indifference or uncertainty about the app's value or effectiveness.

Unlikely to Recommend: A very small group (3.95%) expressed that they are unlikely to recommend the app (rating 2).

"Would you be interested in using an app to purchase discounted, ready-to-go meals from nearby restaurants/eateries?

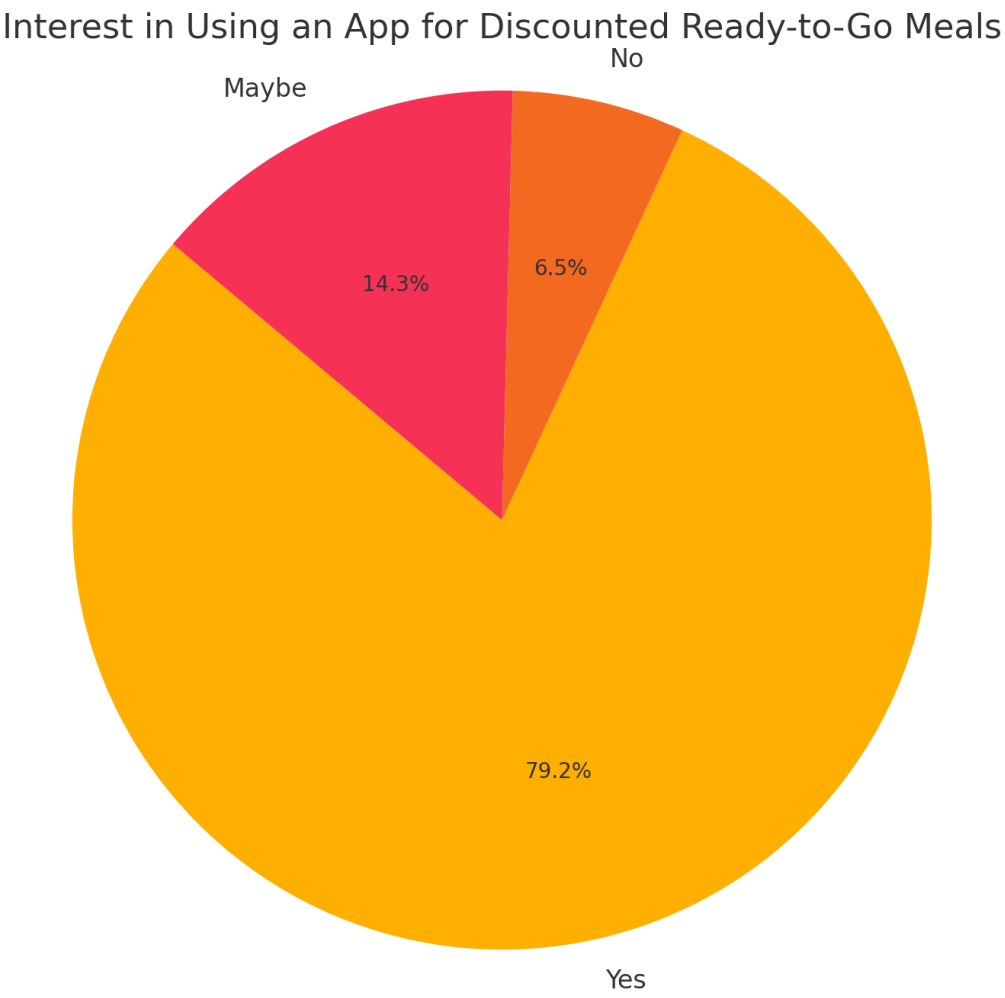


Figure 4.6: Interest in App

Analysis:

Majority Interest: A significant majority, 80%, indicated they would be interested in using such an app, suggesting a strong market potential for this service.

Hesitation: A small portion, 13.33%, responded with "Maybe," indicating some uncertainty or conditional interest that could be due to factors like concerns about restaurant quality, price, or the app's usability.

Rejection: Only 6.67% directly rejected the idea, which is a minimal opposition, but it's crucial to understand their reasons (e.g., preference for cooking at home, dietary restrictions) to address potential barriers in app adoption or marketing strategies.

What benefits would motivate you to participate in a surplus meal app?

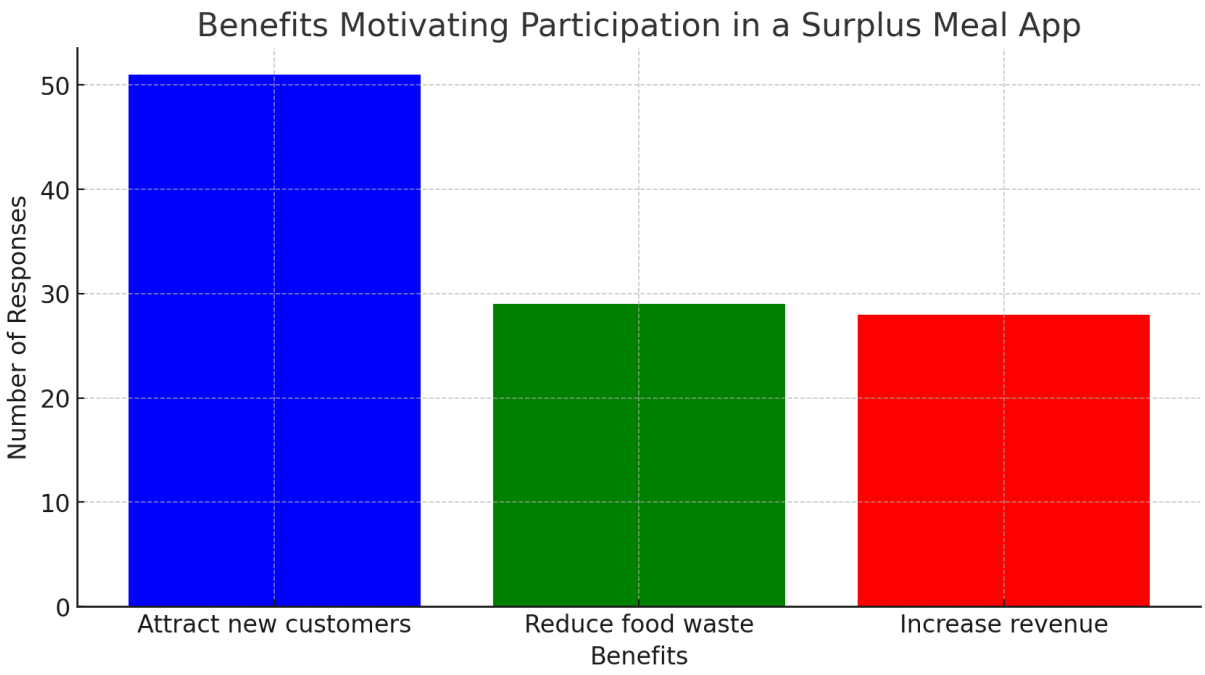


Figure 4.7: Benefits of App Use

Analysis:

Primary Motivator - Attracting New Customers: The majority (67.33%) of the respondents consider attracting new customers the key benefit, highlighting the importance of market expansion and customer base growth for businesses considering participating in a surplus meal app.

Environmental and Financial Incentives: The secondary motivators are reducing food waste (28.71%) and increasing revenue (21.78%). These show significant concern for environmental sustainability and financial gain among the participants. The overlap in these responses suggests that many see a holistic benefit, combining ethical considerations with business growth.

Integrated Benefits: The responses indicating all three benefits together emphasize that participants see a synergistic advantage in using the app, where it not only boosts their customer reach but also enhances their sustainability credentials and financial outcomes.

Concern and Knowledge about Food Waste:

"How concerned are you about food waste in Jordan?" This question gauges the general awareness and sentiment surrounding food waste issues within the local population. By understanding the level of concern among residents, it provides valuable insights into potential market receptivity for solutions aimed at minimizing food waste.

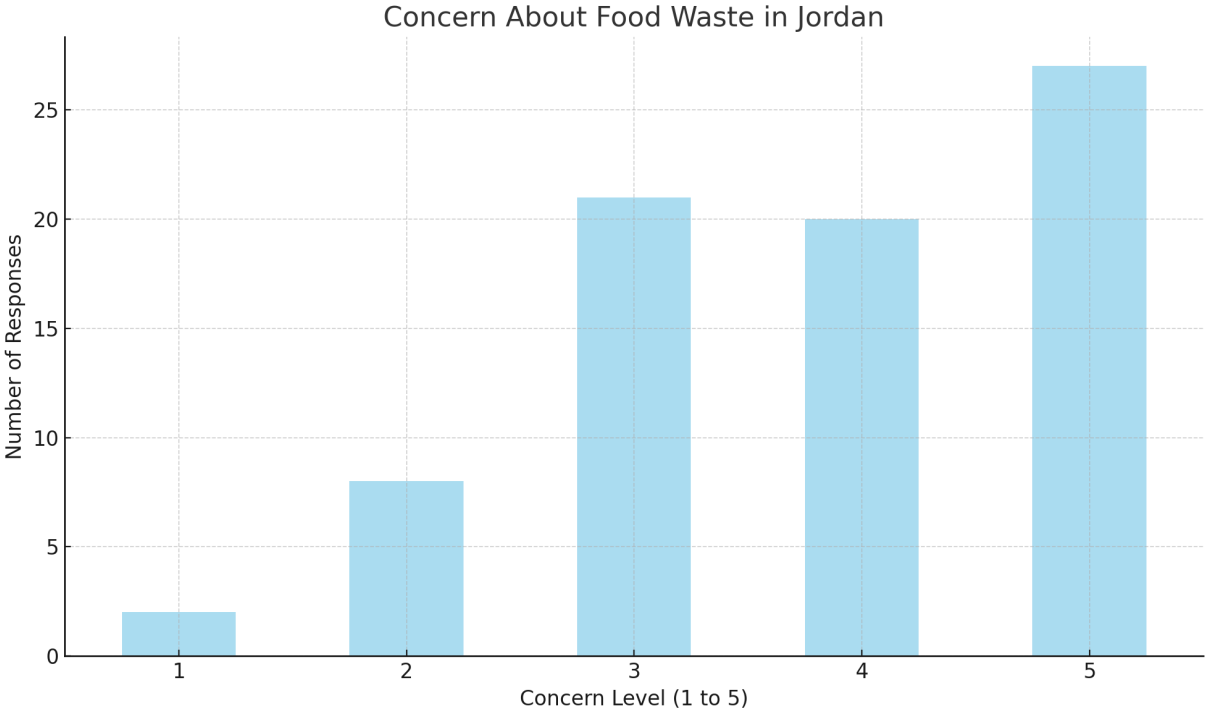


Figure 4.8: Concern About Food Waste Survey

Analysis:

Significant Concern: A substantial percentage of respondents (57.9%) express high to very high concern about food waste (ratings 4 and 5), reflecting significant awareness and concern about this issue in Jordan.

Moderate Concern: A moderate concern level (rating 3) is shown by 27.63% of respondents, suggesting that while they recognize the issue, it may not be a top priority or they feel less impacted by it.

Lower Concern Levels: A minority of respondents express low to very low concern (ratings 1 and 2), making up 14.47% of the total. This could indicate either a lack of awareness or a perception that food waste is less critical compared to other issues.

"Have you noticed an increase in food prices recently?" This inquiry seeks to identify whether rising food costs are impacting consumers' perceptions and behaviors. A noticeable increase in food prices can raise concerns about waste, influencing buying habits and encouraging more efficient use of resources.

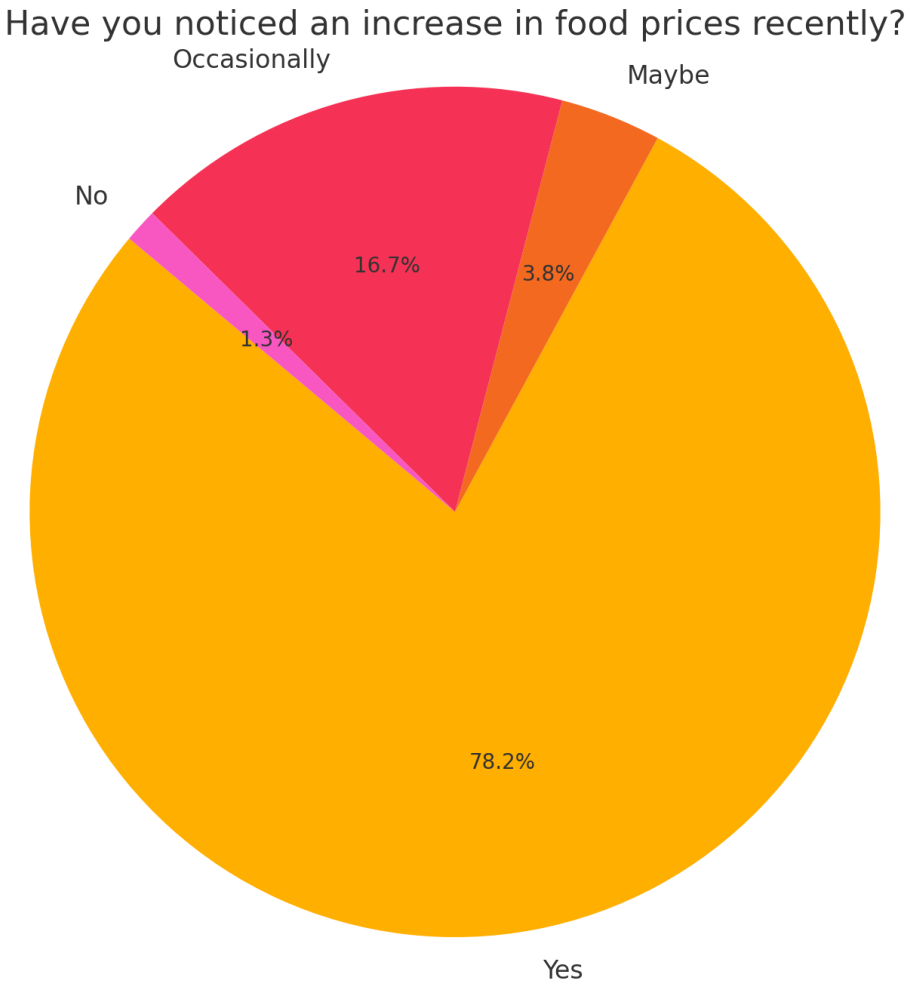


Figure 4.9: Increase in Food Prices Survey

Analysis:

Widespread Perception of Price Increase: A significant majority, 80.26%, affirmatively noticed an increase in food prices, indicating a clear trend that could impact consumer behavior and spending habits.

Occasional Notice: 13.16% of the respondents have occasionally noticed price increases. This response might suggest variability in price changes depending on the type of food, location, or other factors.

Uncertainty: A small percentage (3.95%) responded with "Maybe," showing some uncertainty or lack of clear observation regarding price changes. Negation of Price Increase: Only 2.63% explicitly stated they haven't noticed a price increase, which is minimal compared to the

majority.

"On a scale from 1 to 5, how well do you think this app addresses the issue of food waste?" Here, respondents evaluate how effectively the proposed application mitigates food waste problems. This assessment helps to refine the app's features and strategies, ensuring they align with user expectations and significantly contribute to reducing waste.

Figure 4.10: Effectiveness

Analysis:

High Satisfaction: The majority of the responses are concentrated on the higher end of the scale, with 46.05% rating the app a 4 and 28.95% giving it a 5. This indicates a strong positive perception of the app's effectiveness in addressing food waste.

Moderate Satisfaction: A notable portion, 19.74%, rated the app a 3, suggesting that while they acknowledge some effectiveness, there may be room for improvement or additional features that could enhance its impact.

Low Satisfaction: A small minority (5.27%) rated the app as 1 or 2, indicating significant dissatisfaction or perceived ineffectiveness in tackling food waste.

"What incentives would encourage you to use an app to reduce food waste?" Identifying motivating factors provides a strategic advantage in app design and marketing. Tailored incentives like discounts, rewards, or informative content can effectively encourage widespread adoption and foster a sustainable impact on reducing food waste.

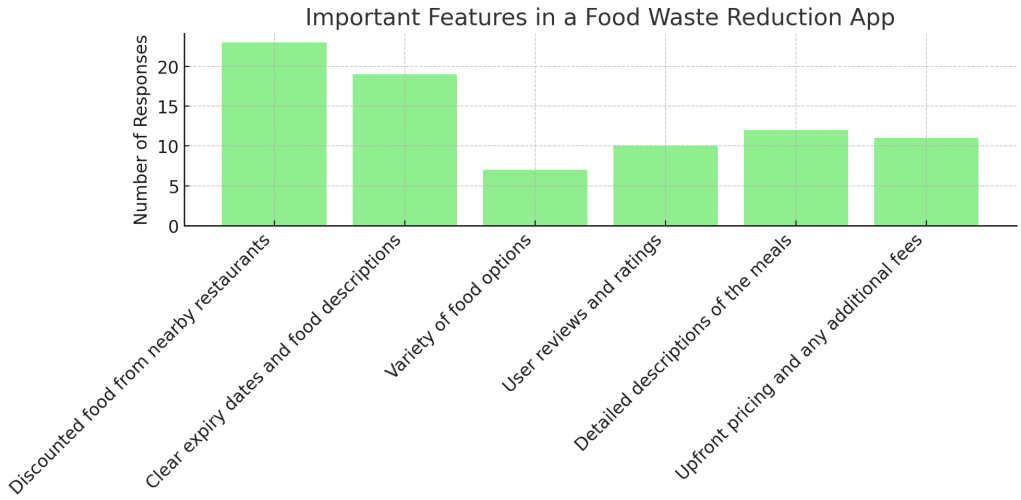


Figure 4.11: Important Features Survey Results

Analysis:

Primary Feature - Discounted Food: The most frequently mentioned feature is discounted food from nearby restaurants, which suggests that cost savings is a significant motivator for app users.

Importance of Information: Clear expiry dates and food descriptions are also highly valued, indicating a need for transparency about the food's quality and safety.

Trust and Choice: User reviews and ratings, along with a variety of food options, highlight the importance of trust in the app's offerings and the desire for diverse choices.

Detail and Pricing: Detailed descriptions of the meals and upfront pricing and additional fees are also crucial, pointing towards a preference for comprehensive information and no hidden costs.

Food Ordering Habits: Questions about the frequency of ordering food from mobile apps, concerns about food waste, and the impact of rising food prices on eating habits fall under this category. They reveal critical insights into respondents' current behaviors, challenges, and attitudes toward food consumption, particularly in relation to the cost and waste of food. What features would be most important to you in a food waste reduction app? (Select all that apply)

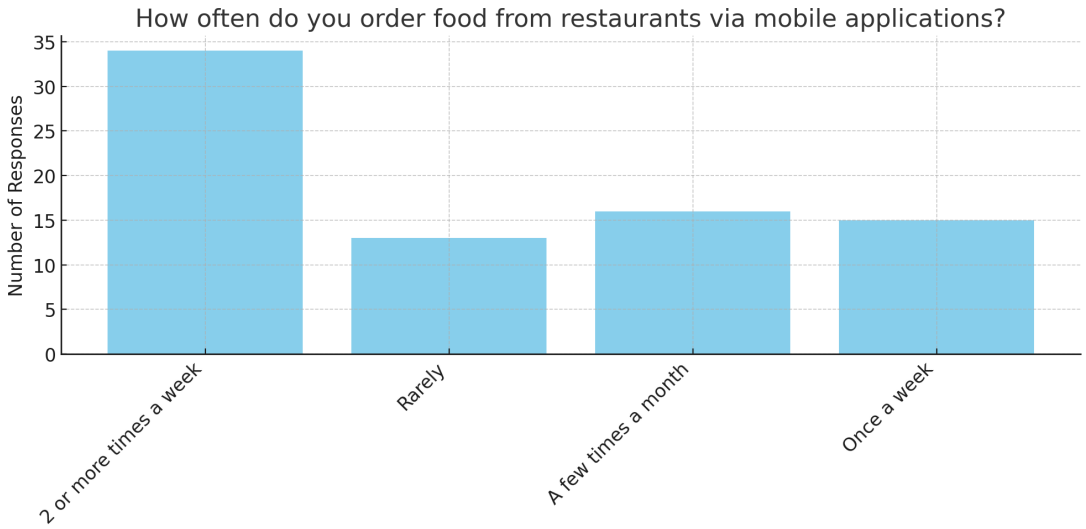


Figure 4.12: Frequency of Ordering Habits Survey

Analysis:

High Frequency Users: The largest segment (46.25%) orders food via mobile apps two or more times a week, indicating a strong habit or preference for the convenience offered by such apps.

Moderate Frequency Users: The next significant segments are those who order a few times a month (25%) and once a week (12.5%), showing a regular but less frequent use, which may depend on specific needs or occasions.

Low Frequency Users: 16.25% of respondents rarely use apps to order food, which might be due to preferences for cooking at home, concerns about costs, or dissatisfaction with available services.

2.2 Qualitative Data Analysis

For the "Karam" app design to match a level of certain consumer needs, the survey presented open-ended questions in an effort to study and enhance the user's experience. Presented in a pyramid structure of close-ended to open-ended questions, this section covers the questions presented to the surveyors.

The surveyors were asked:

1. "Why or why not would you be hesitant to use an app to purchase discounted unsold food? (optional)". For this question, the answers received revolved around the same concern of safety and edibility concerns. The respondents answered "The probability of the food being expired", "False advertising. No monitoring. Does not abide by food and health standards", "Fear of food edibility", and "Food quality, hygiene concerns".

2. "What are your biggest concerns about participating in a surplus meal app?" For this question, the answers varied as follows; "I'm concerned about food safety and maintaining quality, which could impact our brand's reputation if customers view the discounted meals as inferior", "Logistical challenges such as managing pickup times and the additional workload on staff are major concerns", "The pricing structure of the app might negatively affect our regular sales and overall profitability", "I'm unsure about the regulatory and compliance aspects, especially concerning food handling and storage for surplus meals".

3. "What benefits would motivate you to participate in a surplus meal app?". The answers from respondents are as follows: "Reducing waste and enhancing our eco-friendly brand image", "Earning extra revenue from otherwise wasted food.", "Building customer loyalty through community-focused efforts.", "Reaching new, budget-conscious customers.", "Gaining insights from app analytics for better inventory control."

4. "How would you prefer to handle pick-up of surplus meals for customers?" The responses varied as follows: "Scheduled pick-up times during off-peak hours.", "Clearly marked in-store pickup points.", "Option for curbside pickup.", "Using delivery services for convenience.", "Automated pick-up notifications."

5. "Is there anything else you would like to share about your experience with food waste or your interest in this app concept?" Surveryor responses varied as follows: "It's vital the app upholds high standards to protect our brand.", "Curious about system integration with our operations."

Strategic Plan: In this section, students will demonstrate their awareness of the strategic planning process by providing a comprehensive strategic plan for the proposed businesses. The strategic plan is driven from the company description and may include the following: • Systematically surveys and interprets relevant data to understand the business's internal and external environment by applying environmental scanning methods such as SWOT analysis, PESTLE, Porter's Five Forces Analysis, Scenario Planning, Trend Analysis, Benchmarking ... etc. • Set goals and objectives: Based on your environmental scanning analysis, set specific goals and objectives for your company. These goals should be specific, measurable, achievable, relevant, and time-bound

Chapter 5

System Analysis and Design

This chapter provides an explanation of this project's system analysis and design of the application prototype. The below diagrams are used to explain how the system and functionalities of this application would work; Data Flow Diagram (DFD), Use Case Diagram, and an Entity-Relationship Diagram (ER)

1 Data Flow Diagram

A Data Flow Diagram (DFD) is a tool used for a system analyst to construct a visual representation of the flow of data and its activities occurring in a organization/system. The basic elements of a DFD are four elements; arrows, double squares, rectangles with rounded corners, and open-ended rectangles, each representing a part of a system where data flows. The visual representation is composed of processes that display the tasks conducted in a system, data movement is represented by data flow, and data stores hold and save data, while entities communicate within the system (Lucidchart, n.d.). **Instead, the cart's data, such as items added, quantities, and prices, are usually stored in a session or user-specific data structure within the app's broader database system. This allows the cart's contents to be dynamically updated and easily accessed during the user's session.**

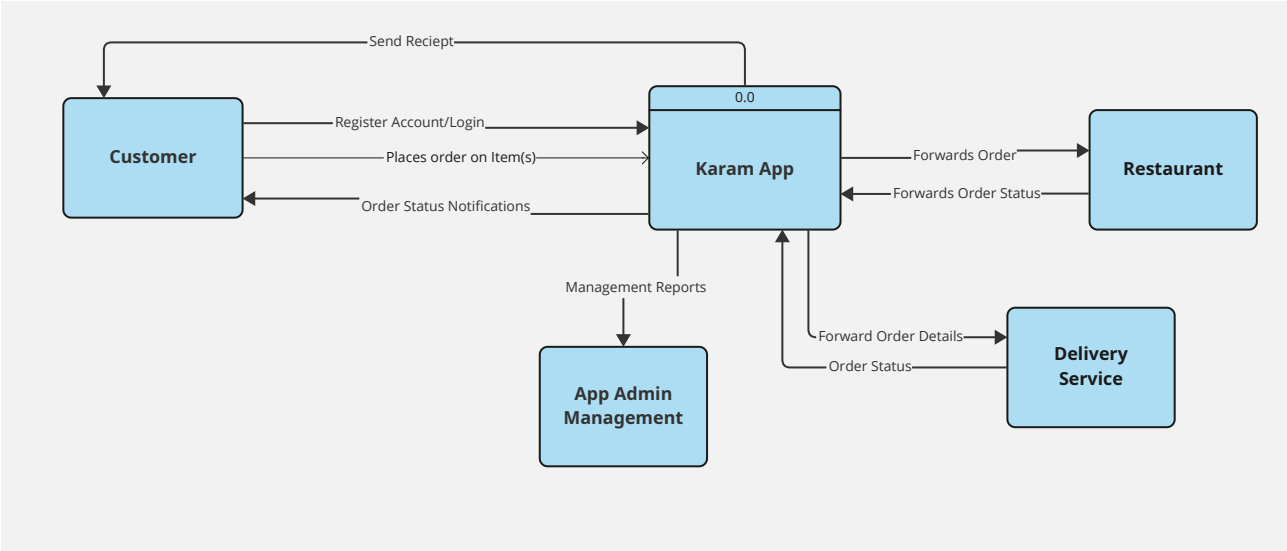


Figure 5.1: Context Level Data Flow Diagram

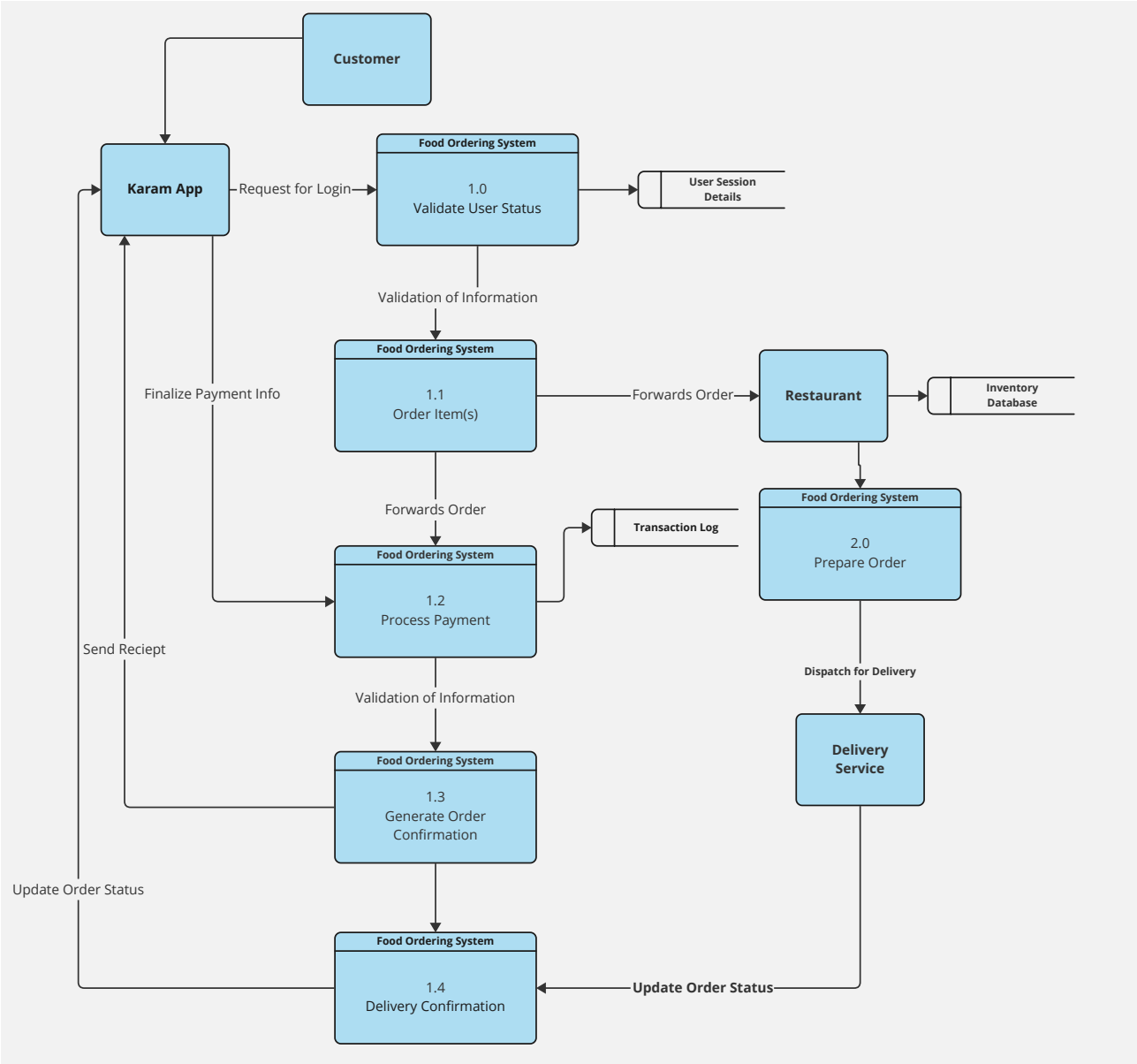


Figure 5.2: Karam Level 1 Data Flow Diagram

The following text will provide a more detailed explanation for the above DFD:

External Entities:

- *Customer*: Interacts with the system to browse and order food.
- *Restaurant*: Provides menu listings and receives orders.

Phases:

- *Item Seeking and Validation:* Customers seek items from a menu, which is cross-referenced with restaurant listings stored in the system. User status is validated, ensuring that actions such as adding items to a cart are restricted to authenticated users.

- *Ordering*: Once validated, customers can add available items to their cart. The session details and user details are tracked to manage the cart and user experience throughout the ordering session.
- Order Validation and Payment: The system validates the information in the cart, including quantities and prices. Payment is processed through a gateway, where transaction details are validated and secured.
- Order Placement and Confirmation: After payment, the order details are finalized and sent to the respective restaurant for preparation and delivery. Confirmation of the order is sent back to the customer, updating their order history and session details for future reference.

Data Stores:

- *Menu List:* Stores the available food items offered by restaurants.
- *User Session Details:* Maintains information about the user's current session, including login status and cart contents.
- *Cart Details:* Holds the items currently in the user's cart, including quantities and prices.
- *Customer Order Record:* A record of completed transactions, used for both customer history and business analytics.

2 Use Case Diagram

Use Case Model is a simulation of how various user types work with the system to address an issue. As a result, it explains user goals, user system interactions, and the activities that the system must take to achieve these goals. This framework is the use case diagram that describes the functions in "Karam" will be performing its primary actors: Customer, restaurant, and Delivery driver.

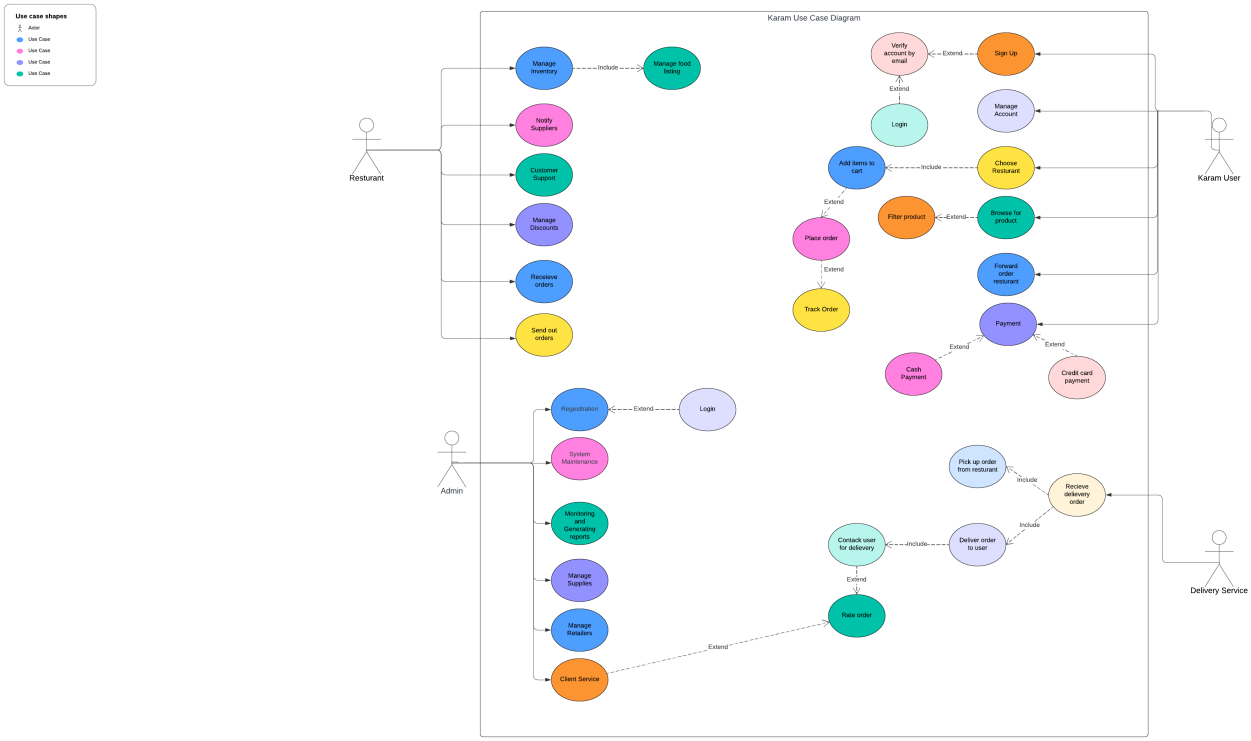


Figure 5.3: Use Case Diagram

Primary Actors:

User: *Actions:*

- *Login:* Users need to authenticate themselves to access their personal accounts.
- *Food Delivery:* Users can browse menus, select food, and place orders for delivery.

Restaurant: *Actions:*

- *Inventory Management:* Restaurants manage the availability of dishes and ingredients.
- *Food Delivery:* Restaurants prepare orders and may handle the initial stage of the delivery process if they coordinate directly with delivery services.

Delivery Service: *Actions:*

- *Food Delivery:* Responsible for picking up food orders from restaurants and delivering them to users.

Secondary Actors

Admin Actions:

- *Login*: Admins log in to access backend systems for user management, content control, and data analytics.
- *Inventory Management:* Oversee and perhaps audit inventory management to ensure data accuracy and compliance with standards.

Actions Detail:

- *Login:* Both users and admins have login functionalities. This action allows for secure access and the ability to customize or administer the platform based on user roles.
- *Food Delivery:* This action is central to the use case and involves multiple actors. Users place orders, restaura nts prepare and hand them off, and delivery services complete the delivery.
- *Inventory Management:* This is primarily a restaurant-focused action with possible oversight or administrative access by an admin. It involves updating and maintaining the stock of food items available for order.

3 Entity Relationship Diagram

An Entity-Relationship Diagram (ERD) is a representation used in database design to visually show the structure and relationships between entities of a system's data model. It consists of entities, which represent existing objects/concepts, represented as rectangles, and attributes which describe the characteristics/properties of these entities. Relationships between entities shown with lines, depict how entities interact with each other.

The relationship types range from one-to-one, one-to-many, or many-to-many, referencing the cardinality and participation constraints between entities. ERD's provide a clear overview of a database's schema, which helps in understanding the system's structure and facilitating communication between stakeholders involved in system design and development.

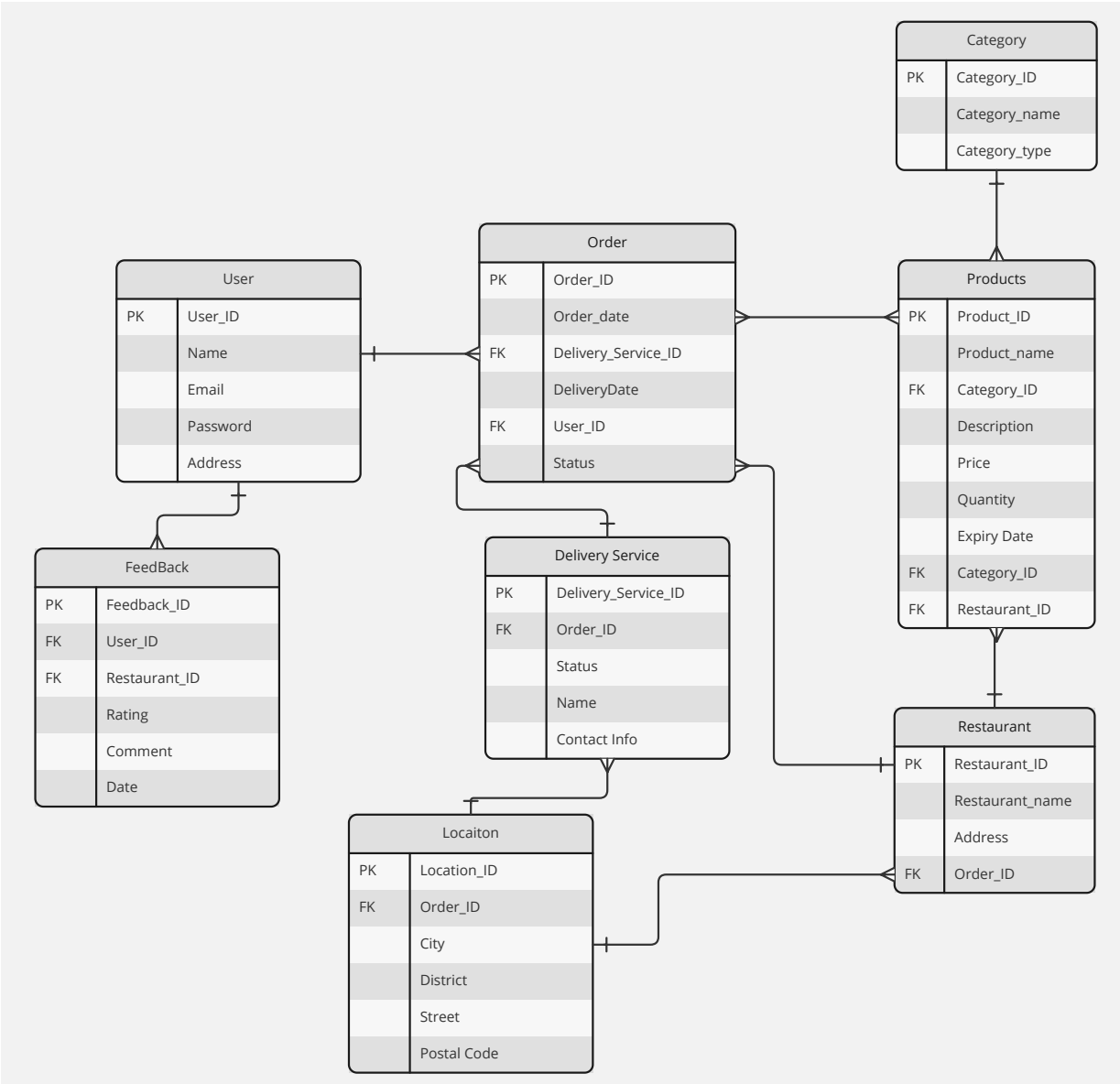


Figure 5.4: Entity Relationship Diagram

- ***User-Order Relationship:*** Users are represented by the User entity which can place multiple orders through the app. This one-to-many relationship captures the action of users interacting with the platform to purchase food items as placing orders. One user can place many Orders as represented by the Orders entity in the above diagram.

- ***Products-Order Relationship:*** Products can be part of multiple orders, and orders can contain multiple products, forming a many-to-many relationship. This relationship captures the essence of the application e-commerce platform, where users can choose from a variety of food items to include in their orders.

- ***Order-Delivery Service Relationship:*** Each order is associated with one delivery service affiliated with the platform, ensuring that each order reaches the user at the specified address. This one-to-many relationship facilitates the smooth delivery process, as the delivery service can be assigned to deliver many orders.

- ***Order-Restaurant Relationship:*** Each order is associated with one Restaurant, indicating that each order placement must be completed with a restaurant. This one-to-many relationship ensures that orders are processed only after placement and the restaurant has accepted and is confirmed.

- ***Restaurant-Product Relationship:*** Restaurants, represented by the Restaurant entity, can offer multiple products for sale on the platform. This one-to-many relationship allows sellers to manage their inventory and reach a wider customer base.

- ***Restaurant-Location Relationship:*** Restaurants have a location assigned in the database for delivery drivers to pick up orders from (forwarded to the delivery service to pick up an order for delivery), creating another one-to-many relationship. This relationship ensures that Restaurants are correctly reached for orders to be delivered.

- ***Category-Product Relationship:*** Products are categorized into specific categories, and each category can contain multiple products. This one-to-many relationship allows users to navigate the platform easily and find relevant products within specific categories.

- ***Location-Delivery Service Relationship:*** Deliveries are made via the delivery service to specific locations, and each location can receive multiple deliveries. This one-to-many relationship ensures that deliveries are directed to the correct addresses.

- ***User-Feedback Relationship:*** Users provide feedback on their experiences with restaurants, forming a one-to-many relationship. This connection allows users to share their opinions and ratings, contributing to the platform's reputation management.

4 Data and Database Design

4.1 User Table:

UserID	Username	Password	Email	UserRole	ContactInfo
001	Anmar_R	123zz	anmar@gmail.com	Customer	07700000

Table 5.1: User Data Table

4.2 Food-Items Table:

FoodID	RestrntID	Description	Quantity	ExpirationDate	Location
S_001	Krispy_K	Donuts	xxx	01/12/2024	Khalda

Table 5.2: Food Items Data Table

4.3 Restaurants Table:

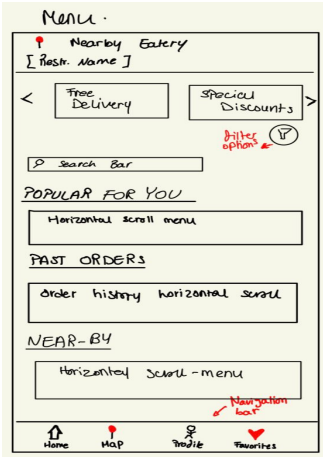
Restaurant_Location_ID	RestrntName	Address	OrderID
S_001	Krispy_K	Donuts	SK344

Table 5.3: Restaurants Data Table

5 Prototyping and UI Design

The below figures shows the link between the concept's intended execution and actual execution, as to provide stakeholders with an initial view of the navigation bar, user interface, and the major features. This stage allows for feedback to be provided and insight into a shared understanding that reflects the final product.

(a) Splash/Welcome Screen



(b) Main Menu Screen

Figure 5.5: UI App Designs

Figure 5.6 (A) above shows the desired design for a home/welcome screen. Users are greeted with a login and create new account options. This page serves the main interface where users enter personal information, a "Login/Signup" page that the user is first greeted with. For a more seamless and convenient user experience, a "Forgot Password?" and Remember Me" options are added. A typical user interface consists of a "Login"/"Sign-Up" form to enter their information, and include "Remember Me" and "Forgot Password" added. Figure 5.6 (B) shows the intended "Main Menu" design for the user interface. The top bar shows the closest eatery to the user based on their precise location, and below that shuffles between options users can choose from app-curated listings, such as "Free Delivery", "New Additions" and such. Below that is a search bar, and restaurants displayed by categories; "Popular For You", "Near-By".

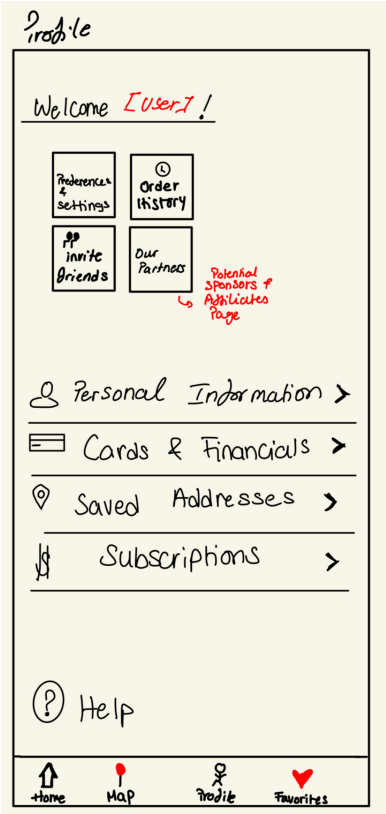


Figure 5.6: User Profile Page

Figure 5.7 shows the "Profile" page the user would change and save their settings within the app. Credit card information, personal details, saved addresses, quick access to a help center line, and subscriptions, in addition to more features can be accessed through this page. Providing the user with an accurate and comprehensive control page enables a friendly interface.

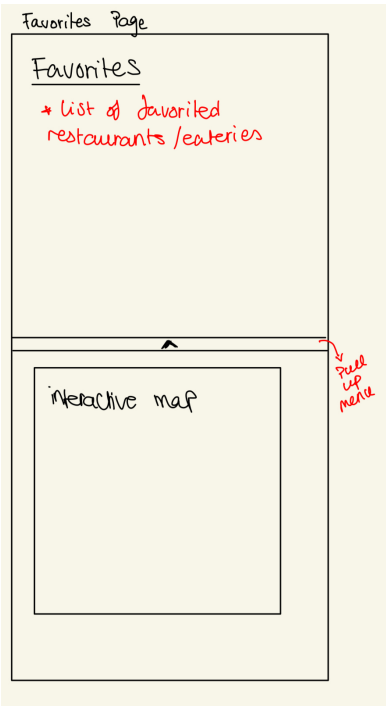


Figure 5.7: Favorites Screen

Figure 5.8 shows the "Favorites" page where restaurants and eateries the user had favorited and saved shows up on this page. An interactive map can be included so users visually see where their favorite restaurants are, and can accurately see where the near-by eateries are.

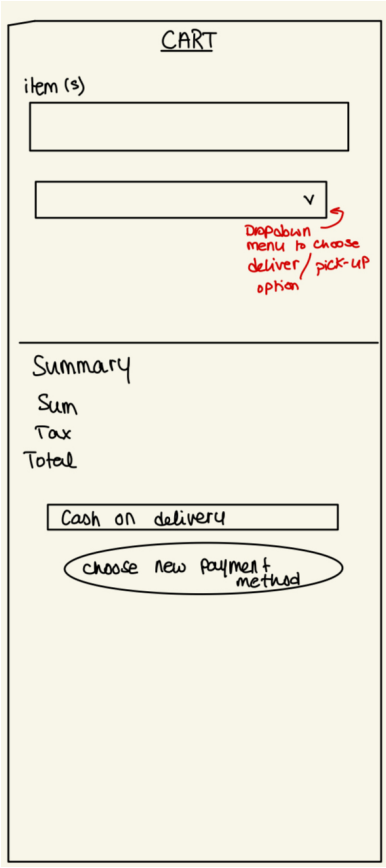


Figure 5.8: Cart Screen

Figure 5.9 above shows the "Cart" page displays the items the users chooses and can choose a method of delivery via a drop-down list, cash or alternative payment method and a displayed summary for item(s) checkout. This page is essential to be seamless for the user.

Chapter 6

Implementation

Chapter 7

Limitations and Future Work

1 Limitations

- *Cultural Aspect:* A sensitivity around dated food could create a social stigma, limiting participation of certain consumers.
- *Technology Barrier:* A segment of Jordan's population might not own a smartphone, or have poor internet reliability and access, which could hinder the reach to a certain demographic. Additionally, digital literacy might be a barrier for some individuals.
- *Supply Aspect:* Businesses could be hesitant to participate due to an unfamiliar business model and logistics. Moreover, concerns over food safety require clearly established protocols throughout collection, storage, and distribution.
- *Demand Aspect:* Logistically, proper chain management and timely delivery requires a network of well-organized logistics. As an added measure, upon signing a new client with the app, restaurants should require a minimum amount of products to be displayed on their menu to ensure user engagement.
- *Poor Infrastructure:* Inaccurate addresses provided through GPS and inaccurate. Infrastructure which is focused on relaying the software system, not physical deliveries. Delivering goods (like unsold food) would require partnering with local logistics companies in Jordan.

2 Future Work

1. *User Behavior and Market Dynamics:* This could be done through examining how the app has affected consumer behavior through looking into things like a increase in the desire of purchasing like "Surprise bundles" or other discounted goods as it can give us an idea to how customers within the app operate and which is used most within the application. Analyzing how the app influences market dynamics within the food sector of Amman, especially the relation to pricing tactics and how competitive reactions from companies that aren't involved within the process.
2. *Scalability and Expansion:* One of the major investigations on expansion for the "Karam" application comes in two possibilities which are to expand locally withing other areas in Jordan and globally with the addition of venturing out in these operations as intended, or

with the intention of adding our operations within the existing delivery app that is "Careem" or "Talabat" as it can aid us extensively in developing the app and adding features easily. Furthermore, it can be considered to potentially adapt other food services within the application such as adding supermarket goods that are prepackaged like (sandwiches, sushi, salads etc.). Through this we can add many different food producers to the Karam app which can increase operations and develop a larger customer base that'll serve many.

3. *Technology Enhancement:* With technology advancing everyday it's best to add AI and machine learning to our operations as it can benefit the Karam app with predictions of food items that are often unsold as it can help to better target offers and reduce waste as intended. Developing features that could enhance the user experience and engagement could be really beneficiary for the app to decrease food waste in Amman. As mentioned before a personalized notifications bar and a loyalty program on users activity and preferences as this could help with tracking users consumption.

4. *Economic Impact Analysis:* The Karam app can deliver many Economic impacts that could benefit many others that are considered through this app as it can help generate new revenue streams including (delivery drivers, restaurants within the app and employees), moreover Karam app could minimize business costs on top of generating income since more revenues and income could lead to better sustain, finally this could lead to expanding a higher customer market as food security is included and accessibility for people within the gap of lower income areas. One of the major options that could be undergo operations is the study of long-term financial benefits for the users of the app, including savings that were accrued from purchasing discounted food.

5. *Social and Environmental Impact:* A very important factor to establishing Karam was to provide food that could be of benefit to the user with the having price and how nutritional the food is and its best to extend the research to study the broader societal impacts of the app, including it's effect on food security and the accessibility of food in lower income areas that is established within Amman. Helping the environment is one of the main reasons of developing this app so examining the environmental impact of reducing food waste through the app could benefit us as it would be known as to how we contribute to the environment and how much has changed after the implementation of the app. This also could be done with the idea of potentially collaborating with environmental agencies to quantity how much of the waste reduction is being made.

Chapter 8

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